More Daycares, Same Mothers at Work: What a Public Childcare Expansion Really Changed

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Abstract

Women traditionally carry most of the responsibility for childcare and household work, which limits their participation in the labor market. This paper studies the effects of a large expansion of public childcare in Chile on mothers' labor market outcomes, childcare choices, and time allocation. Starting in 2007, Chile's main public childcare provider, JUNJI, expanded its facilities across the country. By 2024, facilities opened since 2007 accounted for two-thirds of JUNJI's total enrollment. I exploit variation in children's ages to define treatment and control groups and estimate a difference-in-differences model. I find that the expansion increased JUNJI enrollment but did not affect mothers' employment, while it shifted working mothers toward informal jobs. I investigate two potential mechanisms behind these results. First, mothers may reallocate time toward household work rather than paid employment. Second, mothers substitute JUNJI for other childcare providers rather than increase their overall use of childcare. These mechanisms highlight the importance of understanding families' childcare choices and time preferences when evaluating the labor market effects of public childcare policies.

I Introduction

Mothers still perform most household work and childcare, even when living with male partners, limiting their ability to participate in the labor force. According to the International Labour Organization (2018), women perform 76% of all unpaid care work hours worldwide. Governments have responded to these constraints by expanding access to childcare. Over the last two decades, enrollment in early childhood education has increased substantially. Worldwide, the share of children enrolled in pre-primary education increased from 33% in 2000 to 61% in 2020, while in Latin America it increased from 55% to 76% over the same period (World Bank, 2026).

These policies aim to reduce mothers' childcare responsibilities and give them more flexibility to participate in the labor market (Baker et al., 2008; Halim et al., 2023). However, the evidence on their effectiveness is mixed. Some childcare expansions increase maternal employment (Gelbach, 2002; Berlinski et al., 2011; Nollenberger and Rodríguez-Planas, 2015), while others leave it unchanged (Fitzpatrick, 2010; Havnes and Mogstad, 2011). The literature provides two main explanations for these differences. First, new public childcare places may substitute for arrangements that families were already using, leaving mothers with no additional time (Baker et al., 2008; Havnes and Mogstad, 2011). Second, even when mothers gain time, they may allocate it to household work rather than paid employment, or face labor market conditions that make it difficult to combine work and family responsibilities (Halim et al., 2023).

In this paper, I evaluate Chile's national expansion of free public childcare on mothers' labor market outcomes, and estimate the results mechanisms. I also develop a theoretical model to conceptualize these mechanisms and find behavioral responses. In 2006 the Chilean government launched Chile Crece Contigo program, a system of services that follows children from pregnancy until the age of nine. Its childcare component expanded the supply of the National Board of Early Childhood Education (JUNJI) facilities, the main public provider of daycares. To estimate the effects of this policy, I use the main household survey in Chile to obtain indicators for childcare attendance, labor market, and time use variables. My main sample are mothers between 22 and 45 years old with at least one child between zero and seventeen years old.

To the best of my knowledge, this is the first paper to estimate the causal effects of Chile's childcare expansion using more than 25 years of data and to examine how mothers change their time allocation in response to the policy. I also investigate why the increase in public childcare enrollment does not translate into higher maternal employment. One possible explanation is that families switch between childcare providers instead of increasing their use of childcare. While previous studies have found substitution from informal and private childcare, I find evidence of substitution between public providers. Another explanation is that mothers may use the time gained from childcare for household work rather than paid employment. I also examine whether the employment effects differ between formal and informal jobs. These findings suggest that the effects of childcare policies depend not only on the availability of care, but also on how families respond

to it.

I build a simple two period model of mothers' choices to understand their incentives when facing such a policy. In the model I focus on a representative mother that chooses a daycare provider, whether she works, and how she divides the rest of her time between her child, household work and leisure. Each provider option has a different cost and quality, and the policy affects the probability of having a JUNJI slot. Further, I create a child penalty [1], where if the mother leaves the labor market in the first period, her wage is lower in the future.

The model predicts the patterns I find in the data. First, JUNJI daycares are cheaper and better than municipal daycares. Every family with a municipal place moves to JUNJI when they get the chance. Families paying for private daycare only move if their income is below a certain threshold. Second, employment increases only for mothers whose wage is between their reservation wage with a JUNJI place and without it, and this range is small when the two public options are similar. At the same time, cheaper care makes a few working mothers stop working, so the net effect on employment is small and its sign is not determined. Third, mothers with wages below their reservation wage can send their child to JUNJI while still staying at home, thus stopping to report childcare as a reason not to work. Fourth, moving a child from home to a daycare frees time for the mother, which goes to household work and leisure, while moving a child from one daycare to another does not free any time.

To do the empirical analysis, I implement a difference-in-difference (DID) approach using child age as a eligibility (treatment) definition at the mothers' level. Specifically, I compare mothers with children of 1-4 years old (*treated*) and mothers with children of 6-9 years old (*control*). The pre-policy period is from 1998 to 2006, and the post-period is from 2007 to 2024. I use a national household survey which is a repeated cross-section data. Since there is evidence that the policy targeted mostly low income families, I run this specification by income quintile groups to study whether there are heterogeneous effects depending on the household income. Lastly, for the labor market participation, I distinguish between formal and informal employment using pension contributions as a proxy for job formality, following a commonly used approach in the Chilean context (OECD, 2022). This distinction allows me to examine whether public daycare expansion affects not only maternal employment but also access to jobs with greater social protection.

I find that the policy increase enrollment in JUNJI daycares. With the survey data I observe that treated women enroll their children 5.2 points more on average relative to what would had happened on the absence of the policy, and by 2024 it is 10.4 points higher. Relative to its own trend before the policy, JUNJI enrollment ends 5.6 children per 100 above its old path in 2024. Furthermore, the new facilities (opened from 2007 onward) enroll 9.8 children per 100 children by 2024, which is two thirds of all JUNJI enrollment that year. The results by income group show that attendance rises 2.6 points for low-income children, 6.6 for middle income children, and 2.6

[1] The child penalty refers to the negative effect of parenthood on women's labor market outcomes relative to men's, particularly in terms of employment, working hours, and earnings (Kleven et al., 2019).

for high-income children. Meaning that middle income families were the most responsive to the policy.

Even the positive and strong results on take-up, I observe that mothers do not improve their labor market outcomes. The results for employment and looking for a job are small and non-significant independent on the controls and sample. For working women, there is a decrease on their hours allocated to work, the effect is around 5% (2 hours on a 38 hours working week). Their monthly income decreases in the same rate as their working hours, hence, their hourly wage could have been maintained equal. In terms of formality in jobs, formal work falls 2.7 percentage points (from 29%) and informal work rises 2.1 (from 15%), so total employment barely changes.

I study two mechanisms of why daycare demand increases but mothers' labor supply did not respond. One is *substitution*, where mothers who were already using a daycare, change providers because JUNJI could be an improvement in quality or in their budget. I show that mothers actually change providers among public supply. By 2024, municipal enrollment is 6.2 below its pre-2006 trend, about half of its 2006 level. The municipal decline offsets all of JUNJI's growth above its trend, and about 60% of the 9.8 children per 100 enrolled in the new facilities.

The second mechanism is mothers' *time allocation* decisions. Among mothers who remained inactive, I show that using childcare constraint as their reason to not look for a job decreases for treated mothers. Compared with mothers who were never exposed to the expansion, inactive mothers with a child of daycare age are 2.9 percentage points less likely to indicate childcare as their reason for being inactive. While, they still are out of the labor force, they state the reason "they need to do household work" to explain their labor force absence. Selecting household work increases 4.5 percentage points, from a base of 33.8%, which is about a 13% increase. In the two-by-two, this shift is concentrated in the poorest households, where the share of mothers citing childcare falls 4.1 percentage points. For the reason, "not interested" there is not evidence that this was reported more, not even across income groups.

I conduct several additional analyses to assess the robustness of the main findings. First, because treatment assignment is based on the child's age, some mothers assigned to the control group may have been exposed to the policy in earlier periods. To address this concern, I estimate treatment effects using the staggered difference-in-differences imputation estimator proposed by Borusyak et al. (2024). Second, I estimate a triple difference using fathers as an additional comparison group. Third, to isolate other parts of the policy I use mothers with children of 6-7 years old as control groups [2]. The main findings are broadly consistent across these alternative approaches. The staggered estimation is presented together with the main results (Section VI.V). In the Appendix, I present the triple difference using fathers (Appendix A), mothers of children aged 6 to 7 as the control group (Appendix B), the results without the 2006 survey wave (Appendix C), and the

[2]The CCC program offers an after school policy for children of 6-12 years old. The number of children (families) affected by this policy is small. There is also another policy, outside of this program, called "full time school," affecting all children between 8-12 years old.

results comparing women inside the same level of local JUNJI supply (Appendix D), followed by the fertility estimations, the derivations of the model, and the distance from households to JUNJI daycares.

This paper relates to three literatures. The first one studies how childcare policies affect mothers' labor market outcomes, and I contribute by explaining why a large public expansion did not increase employment. Some papers find increases in maternal employment (Gelbach, 2002; Baker et al., 2008; Berlinski et al., 2011; Bauernschuster and Schlotter, 2015; Nollenberger and Rodríguez-Planas, 2015), while others find small or no effects (Fitzpatrick, 2010; Havnes and Mogstad, 2011). Evidence from developing countries also points to employment gains, although their size depends on the setting and the services offered (Halim et al., 2023). In Chile, Encina and Martínez (2009) and Aguirre (2015) find limited employment responses to childcare for young children, while Martínez and Perticará (2017) find gains from afterschool care for older children. I evaluate a national expansion using 26 years of data, 18 of them after the policy, and I connect the null result to substitution across providers and to the reasons mothers give for staying at home. Being the first paper in Chile to causally estimate the effects of this policy and test the mechanisms.

The second literature studies how mothers divide their time between childcare, household work and paid work (Kimmel and Connelly, 2007). These activities can respond differently to childcare policies. For example, Lee (2016) finds that a universal childcare subsidy in Korea increased the use of childcare centers and reduced maternal care, with little effect on mothers' labor supply. In Chile, cultural factors also help explain women's labor force participation (Contreras and Plaza, 2010). I contribute by showing that after the expansion, mothers of the youngest children report household work more often as their reason for not looking for a job, while their employment does not change. Childcare can become less important as a reported barrier while household responsibilities still keep mothers at home. To the best of my knowledge, this is the first paper to estimate how a childcare expansion changes the reasons mothers give for stay out of the labor force.

The third literature studies how public childcare expansions substitute for existing care arrangements. Previous work looks at substitution between public and private provision (Bassok et al., 2014), between formal and informal care (Havnes and Mogstad, 2011; Cascio, 2009), and between public programs (Bassok, 2012). I contribute evidence on substitution between two public providers. JUNJI enrollment increases while municipal enrollment falls below its pre-policy trend, and in the model this movement comes from the differences in cost and quality between the two providers. This distinction matters for policymakers, because an expansion can change where children receive care without a similar increase in total coverage.

The paper continues as follows. Section II presents the model, Section III describes the policy and the Chilean context. Section IV presents the data, the sample and the outcomes. Section V explains the empirical strategy. Section VI presents the results, and Section VII concludes. The Appendix presents the robustness checks and the model derivations.

II A Model of Mothers' Choices

I develop a simple model to explain why expanding public childcare can increase enrollment without increasing maternal employment. The model distinguishes three margins: the choice of childcare provider, labor force participation, and the allocation of non-labor time. The model shows that increased access to public childcare does not necessarily remove the barriers that prevent mothers from working. Mothers may substitute between providers, or use newly available care while remaining out of the labor force. The model organizes the empirical findings rather than providing a framework for estimation.

II.I Environment and choices

My model focus on a representative mother [3] with only one child, observed over two periods. In each period, the mother chooses a daycare provider j_t, employment $h_t \in [0, \bar{h}]$, consumption x_t, leisure l_t, household work z_t, and time with her child b_t. She solves

$$\max \; u_1 + \beta \mathbb{E}_1[u_2], \qquad u_t = \ln x_t + \phi_l \ln l_t + \phi_z \ln z_t + \phi_b \ln b_t + \phi_q \ln q_t, \tag{1}$$

where the preference weights, ϕ_j, are positive. q_t is child development, which combines time with the mother and quality-adjusted daycare:

$$q_t = \left[\alpha_t(\theta_{j_t}\bar{a})^\rho + (1-\alpha_t)b_t^\rho\right]^{1/\rho}, \qquad 0<\rho<1, \quad 0<\alpha_t<1. \tag{2}$$

Where $\bar{a}$ is the maximum amount of hours in daycare, and θ_j denotes the daycare quality (varying across providers). Maternal time therefore enters utility directly and through child development. The daycare input (first terms in equation 2) is zero when the mother provides care at home.

The mother faces the following budget constraint (which is binding for simplicity),

$$x_t = y - p_{j_t} + w_t h_t \tag{3}$$

where x_t is her private consumption at year t, and p_j is daycare cost [4]. y is non-labor income (e.g., spouse's income), and w_t is her hourly wage. She also faces a time constraint,

[3] Future work is to expand this model into heterogeneous mothers with different level of education. Hence their childcare preferences differ by their observed constraints but also by their preferences of childcare quality. Where, my assumption is that more educated mothers will be willing to pay more for the same increase in daycare quality.

[4] The price of the private good is normalized to 1 to simplify the model solution.

$$L = h_t + l_t + z_t + b_t, \tag{4}$$

$$\bar{a} + b_t = S \quad \text{if } j_t \neq H, \qquad h_t = 0 \quad \text{if } j_t = H. \tag{5}$$

L is the mother's time endowment divided in working hours, h, leisure, l, household work z, and motherhood b. The child needs to be cared for a total of S amount of hours, which has to be equal to the time in daycare, a, and the time being cared at home, b. There is a direct relation between the hours of work and childcare at home. Meaning, the child cannot be left alone, assuming the mother is the only caregiver at home, she cannot work more than the hours supplied by the chosen daycare, $h \leq \bar{a}$.

Care options are maternal care at home (H), informal care by a relative (I), municipal daycare (M), JUNJI (J), and private daycare (P). I assume

$$p_H = p_I = 0 < p_J < p_M < p_P, \qquad \theta_I = \theta_H < \theta_M < \theta_J < \theta_P. \tag{6}$$

where p_j are prices, and θ_j represents provider's quality. I assume that public daycares have a small price which is slightly larger than zero. For instance, for public providers, p_j represents access expenses such as transport rather than tuition. These rankings are maintained modeling assumptions: JUNJI dominates municipal care in price and quality, while informal and private care involve price–quality trade-offs. Home, municipal, and private care are always available; informal care requires an available relative.

A JUNJI place is offered with probability $\pi_{J,t}$. The policy is produces an increase in $\pi_{J,2}$, leaving first-period choices unchanged (no-anticipation).

Employment in the first period raises the second-period wage:

$$w_2 = \underline{w} + \delta \, \mathbf{1}\{h_1 = \bar{h}\}, \qquad \delta > 0. \tag{7}$$

This captures the earnings cost of an employment interruption. Mothers who previously remained at home consequently face lower wages when childcare expands.

II.II Care, employment, and time allocation

I assume that the maternal-time constraint binds under external care, so $b_t = \bar{b} \equiv S - \bar{a}$, and that mothers providing care at home choose $b^H > \bar{b}$. I also assume that working with external care leaves less time for leisure and household work than remaining at home. These restrictions isolate the trade-off between maternal time, care quality, and market work. Appendix F provides the conditional allocations and choice thresholds.

Consider the second period and suppress time subscripts. Let $V_j(h)$ denote utility after optimizing time allocation conditional on care j and employment h. The mother chooses the highest-value feasible pair. For a given external provider, the reservation wage is

$$w_j^* = \frac{(y - p_j)(e^{\Gamma} - 1)}{\bar{h}}, \qquad \Gamma = (\phi_l + \phi_z) \ln\left(\frac{L - \bar{b}}{L - \bar{h} - \bar{b}}\right). \tag{8}$$

She works with provider j when $w \geq w_j^*$. The term Γ measures the utility cost of reallocating leisure and household work to employment. Higher nonlabor income raises this threshold. Conditional on the same provider, care quality does not affect the work decision because care hours, maternal time, and child development are identical under employment and nonemployment. Quality nevertheless affects provider choice and the comparison between working with external care and remaining at home.

This distinction generates three implications for the childcare expansion.

Enrollment and substitution. Greater access increases JUNJI enrollment among mothers who prefer it to their existing options. Municipal users switch whenever offered a JUNJI place because JUNJI is cheaper and higher quality.

For a mother with resources $m = y + wh$ and the same employment status, the gain of moving from provider k to JUNJI is

$$V_J(h) - V_k(h) = \ln\left(\frac{m - p_J}{m - p_k}\right) + \phi_q \ln\left(\frac{\bar{q}_J}{\bar{q}_k}\right), \qquad \bar{q}_j = \left[\alpha(\theta_j \bar{a})^{\rho} + (1 - \alpha)\bar{b}^{\rho}\right]^{1/\rho}. \tag{9}$$

where m is the household's resources before paying for childcare: non-labor income plus the mother's earnings, p_j are daycare prices. And q_j is child development conditional on choosing a daycare.

For municipal users both terms are positive, since $p_J < p_M$ and $\theta_J > \theta_M$, so they always switch. For informal users the first term is negative and the second positive, and for private users it is the opposite, so private users switch only if m is below the threshold $\bar{m}$ that makes equation (9) equal to zero.

Informal users switch when the quality gain outweighs the access cost; private users switch when the savings outweigh the quality loss. At a fixed employment status, higher resources make the quality premium of private care more affordable. Thus, enrollment growth can reflect substitution between providers rather than a reduction in maternal caregiving. Equal access expansions need not produce equal enrollment responses across locations, because the share preferring JUNJI may differ.

Employment. The net employment effect is ambiguous.

When the probability of a JUNJI place increases by $\Delta\pi_J$, the change in employment is

$$\Delta E = \Delta\pi_J \Big[\underbrace{\mu_H \Pr(\max\{w_J^{**}, w_J^*\} \le w < w_M^{**}) + \mu_I \Pr(w_J^* \le w < w_I^*)}_{\text{entry}} - \underbrace{\mu_M \Pr(w_M^* \le w < w_J^*)}_{\text{exit}} \Big], \tag{10}$$

where μ_k is the share of mothers initially in option k, w_j^* is the reservation wage of equation (8), and w_j^{**} is the wage at which working with provider j is as good as staying at home with the child. The exit interval has width $w_J^* - w_M^* = (p_M - p_J)(e^{\Gamma} - 1)/\bar{h}$, and both the entry and exit intervals shrink as JUNJI and municipal care become similar in price and quality, so the sign of ΔE is not determined and its size can be small.

A mother initially providing care at home may enter employment if JUNJI raises the value of combining work and external care sufficiently. However, she can also enroll her child and remain out of work if her wage is below the relevant threshold. Mothers already using external care gain no additional time from switching providers. For municipal users, lower care expenses raise disposable resources and increase the conditional reservation wage, potentially inducing exits. For informal users, the positive cost of JUNJI can instead encourage employment. The aggregate effect depends on the initial care distribution and the mass of mothers near the relevant choice thresholds. A small effect is therefore possible, but is not guaranteed by the model. Lower wages following an earlier employment interruption can further limit entry among previously nonworking mothers.

Nonmarket time and reported reasons. A mother who moves her child from home to JUNJI without entering employment reallocates time toward household work and leisure. In particular,

$$z^0 - z^H = \frac{\phi_z}{\phi_l + \phi_z}(b^H - \bar{b}) > 0, \tag{11}$$

with an analogous expression for leisure. Switching between external providers leaves time allocation unchanged at a fixed employment status.

To connect these choices to the survey, I impose a reporting rule: a nonworking mother providing care at home reports childcare as her reason for not seeking work; a nonworking mother using external care reports household chores or no interest in working. A home-to-JUNJI transition without employment therefore changes the reported reason without changing employment. This is a mapping from choices to survey answers, rather than a separate prediction about preferences. Aggregate reporting changes also depend on employment transitions. Moreover, reported chores do not directly measure hours of household work.

The model reconciles increased JUNJI enrollment with provider substitution, limited employment responses, and changes in reported reasons for nonparticipation. It separates improvements in care price or quality from increases in available maternal time, and distinguishes both from the incentives to undertake paid work.

III Policy Intervention

The National Employment Survey estimates that in 2022, 25% of people who did not look for a job, or who were not able to work, were constrained by *permanent family reasons*.[5] Out of this group, more than 95% were women.[6] This, along with other concerning evidence of gender and social inequality (Martínez and Perticará (2017)), prompted President Michelle Bachelet to launch the Chile Crece Contigo (Chile Grows with You) program in 2006.

This program offers services for children and their families, with additional targeted services for families in vulnerable situations. A woman (family) usually enters the program at her first prenatal checkup in a public health center, where a midwife follows the pregnancy, and a child can enter at the first well-child checkup with a nurse. The program follows the child until the age of nine.

In this paper, I focus on the education layer of the program. Specifically, the program aims to secure access to public childcare for every family that needs it, together with the "Junta Nacional de Jardines Infantiles" [7] (JUNJI from here on) institution as the main supplier. The program increased the availability of slots (i.e., increasing the supply of public daycares by creating more facilities or expanding current ones). Other institutions such as INTEGRA and Municipalities help into the provision of public daycares, but with a smaller share compared to JUNJI. Figure 1 shows a first glance of the geographic change of this policy in terms of creating new facilities.

Figure 2 shows the number of JUNJI childcare centers, slots, and enrollment at a national level between 2000 and 2025. This figure uses all the cities in the administrative data and not only the ones in my sample, so it gives the national picture. We can observe a sharp increase in 2008 for capacity and enrollment, while a smoother increase in facilities starting in 2006. This makes sense since according to the policy they wanted to increase the provision of slots, which could be done by expanding current facilities and constructing new ones. Even though the policy started in 2006, the expansion is observed a few years later. This could be due to all the work that needs to be done for this expansions (e.g., health and education department approvals, new staff hires, resources allocation decisions, among others). These figures tell us that the timing of the policy was indeed significant and it is the better candidate to indicate before and after.

One thing interesting to note, which it is a small contribution in terms of policy and data construction is understand the type of programs JUNJI offers. As explained in more detailed in the next section, JUNJI data is provided by its data center administration. However, even if this is a common data source for researchers, it is important to be careful on what we treat as a daycare and as a facility to free mothers' or caregivers' responsibilities. There are six type of programs that JUNJI offers to children (and their families) under 5 years old. Two of them, are not included

[5] According to the survey, the definition of "Permanent family reasons" correspond to family responsibilities such as children or any other dependent care.

[6] This number corresponds to the third quarter of 2022.

[7] the translation is National Board of Early Childhood Education

Figure 1: Heat map of Chile: JUNJI facilities per 1,000 low-income children (2000, 2025)

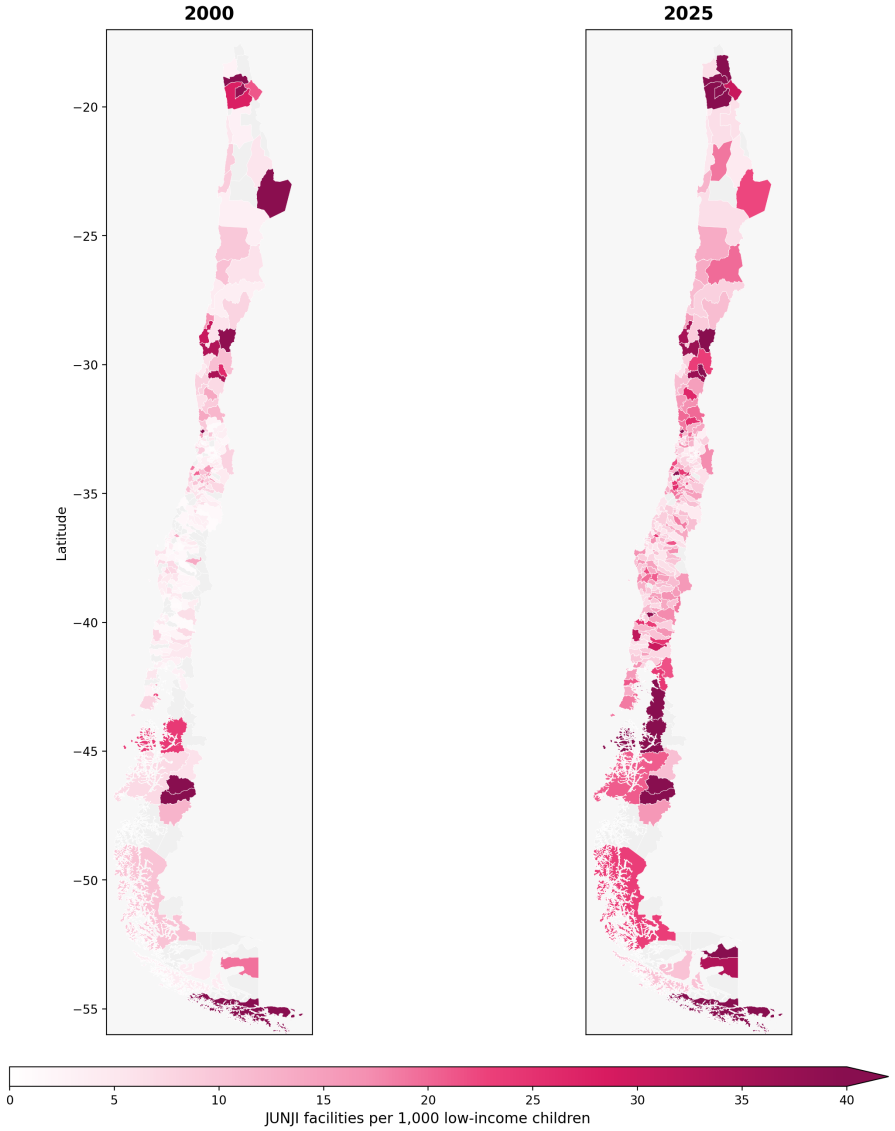
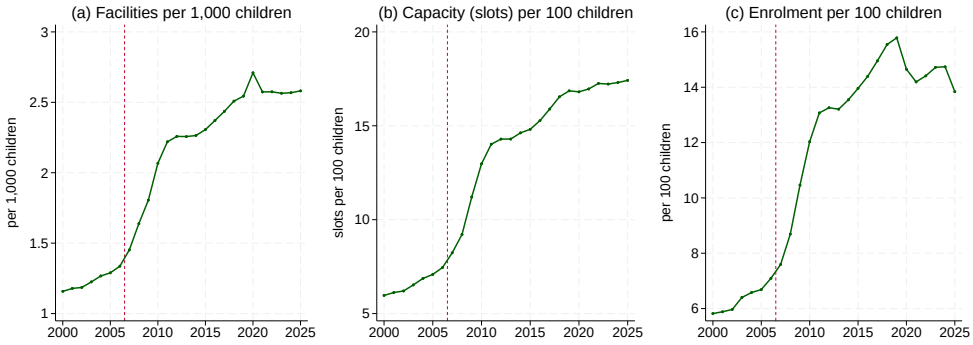
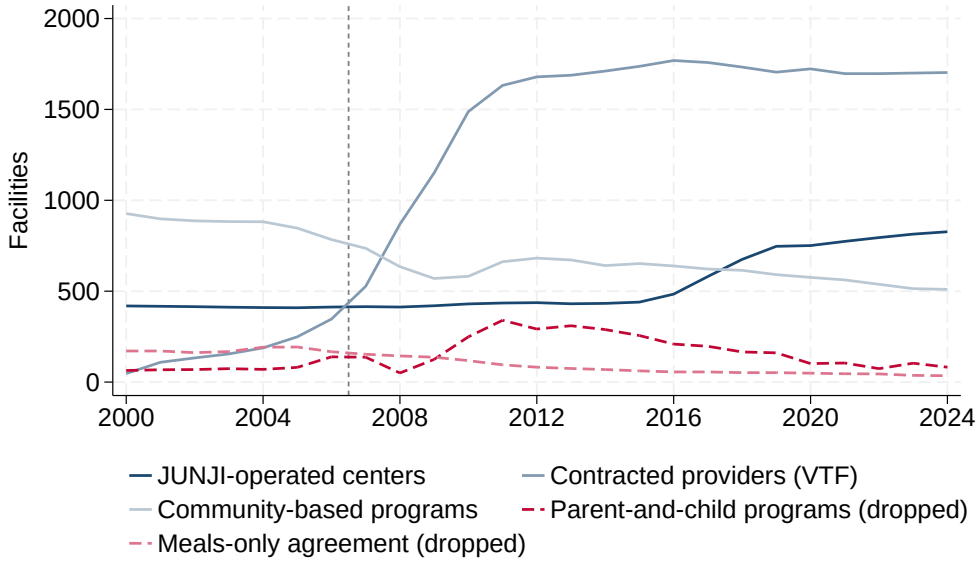


Figure 2: The JUNJI expansion



in this paper, since they do not accomplish the goal of free mothers' time and responsibilities of childcare. One is named "parent-and-child" program, which indeed requires a parent (or guardian) to go to daycare with their child. This program represent close to a 10% of facilities on the national JUNJI provision (pre policy, 2006). The other is called "Meals-only agreement ," this program that appears as a facility in the data only give children already enrolled in another childcare the possibility and access to meals (breakfast, lunch, and dinner). So, again, this does not fit the job I am looking for a childcare facility for the purpose of this paper. Figure 2, in panel (a) shows the national expansion for all JUNJI daycare facilities. While, Figure 3 shows the increase of the different JUNJI facility programs between 2000 and 2025. You can note that the main JUNJI increase is driven by contract providers (VTF, because of its meaning in Spanish). This daycares are financed and ruled by JUNJI but managed by a third party, usually municipalities.

Figure 3: JUNJI expansion by program



One thing to rule out is whether this increase in public supply of daycares was because of an shock that positively affected all providers (or at least public providers). If that were true, then we should observe an increase in the number of daycares coming from any provider. Figure 4 shows that this is not the case. The three other groups that are important in the Chilean context, private, INTEGRA, and municipal, actually show flat curves. Voucher is a special case where shows an inverted U-shape. It is out of the scope of this paper to explore this behavior.

Figure 4: Number of childcare facilities by provider

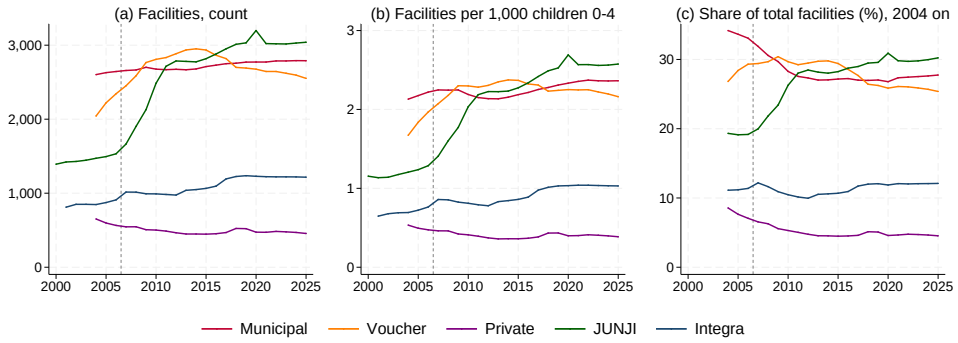
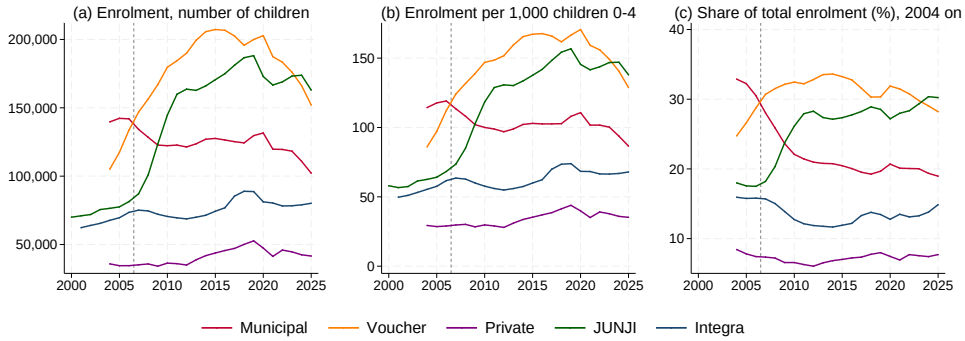


Figure 5: Enrollment and facilities by provider: counts, rates and shares



IV Data

In this paper I use data from five main sources. The main source is the Chilean household survey called "CASEN". This survey is a repeated cross-section that started in 1987. I am using it since 1998 until 2024 with two-three years gap. This survey covers approximately 69,000 households, and a little over 200,000 individuals per year, which makes it representative at the national level. I use this data for labor market outcomes, household composition, income analysis, and childcare attendance.

The second source is where I obtained the number of JUNJI and INTEGRA facilities, enrollment, and capacity. I use the "Transparency Law" in Chile that requires public institutions to send data to any user that requested it. I have this data at the facility level from 1998-2025. Third, to account for the number of facilities and enrollment in other type of daycares I use the ministry of education data portal, where they have childcare population records from 2004 onward at the city-year level.

To measure enrollment and facilities in "real" terms, I use the population estimates and projections by comuna that INE builds from the 2017 Census, which give the population at the city, age and year level. These are projections and not an observed count for every year, since a census is

only taken every ten years. From there, I am able to account for the number of children under 5 years old in each city for every year in my study. This allows me to use facilities per 1,000 children, and enrollment per 100 children as measures of daycare supply and demand at the city-year level, respectively.

I also use the National Employment Survey (ENE), from the National Institute of Statistics, which is where the numbers on permanent family reasons of the previous section come from. I use the ENE only to compare and to describe some labor statistics to ensure the use of the CASEN for it. It does not identify cities in the way I need for the econometrics analysis, so I do not run any regression with it.

I build the city sample from the cities where JUNJI had some presence during the pre-policy window, 2000 to 2005, in any of its programs. This leaves me with 322 cities out of the 335 that appear in the administrative records.

My main sample are mothers between 22 and 45 years old who live with at least one child that is zero to seventeen years old. I leave out older women, because at those ages they are more probable to be grandmothers than mothers, something that is difficult to control for with the available data. I have 321,158 women, and 528,077 children in the entire sample. I use the age of the child as the main treatment assignment. Following the common differentiation of *eligible* and *non-eligible* groups to estimate DID (or RCT in some cases). Hence, mothers with children of 1-4 years old are eligible (*treated*, 119,028 women), since their children can attend JUNJI daycares. While mothers with children of 6-9 years old are non-eligible, belonging to the *control* group (81,733), since those children are not able to attend a daycare. Mothers with newborns and/or children older than nine years old are use for the imputation staggered DID estimation presented in the next section.

IV.I Outcomes

I estimate several outcomes, and I divide them in four groups which will be represented separately in the Result section. First, *take-up*, I use a dummy variable indicating whether the child goes to a JUNJI daycare (I also check attendance to any public daycare [8]). I complement it with the administrative records of JUNJI, which report the average enrollment of every JUNJI facility in each year, by program and level. With them I build, for each city and year, the JUNJI enrollment per 100 children between 0 and 4 years old. I also separate the enrollment in the facilities that opened in 2007 or later from the enrollment in the facilities that already existed.

Second, *labor market outcomes*, these are at the mothers' level using survey data, I have two dummy variables, one indicating whether the mother is working or not, and another to indicate whether the mother (conditional on not working) is looking for a job. I also have the log of income and weekly working hours, the latter to measure the intensive margin of working, and two dummies

[8] I use this variable to relax a possible measurement error coming from parents' incorrectly answering where their children go to daycare. It is easy to mistake daycare providers among public supply.

for formal and informal work.

Third, *substitution* effects, here I am concentrating on enrollment in other type of daycares (municipal, private, and voucher). I use indicator variables from the survey for attendance to any of this providers (separately). And, I use the enrollment registers of the Ministry of Education (MINEDUC) to build the municipal and private enrollment per 100 children between 0 and 4 years old. The idea is to be able to observe whether there is change of preference across providers from the working-mother perspective.

Fourth, I use variables representing mothers' *time use*. Specifically, I use three dummy variables that I create from the answer to the question: "Why you did not look for a job," so we can assume that these mothers are inactive. The variables indicate three reasons (separately), one is "because of childcare constraint", then, "because they have to do housework," and "they are not interested." I measure them as shares of all women and of inactive women. The idea for this ones is to represent the use of time for inactive mothers, and their outside options for not being working.

IV.II Descriptive Statistics

Table 1 reports the outcome variables listed above, separately for the three levels at which I estimate. Panel A is children aged 1 to 4 (6 to 9 for the control group), panel B their mothers, and panel C the cities. The columns are the pre-policy and post-policy means, so the table also states the sample sizes behind each set of results. Attendance rises across every public (and voucher) provider between the two periods, while private attendance is flat. Treated mothers' employment goes from 42% to 54% (12pp). While, employment for mothers in the control group goes form 48% to 61% (13pp). Most of the increase in mothers' employment comes from formal jobs, which rise from 26.7% to 38.1% for treated mothers and from 29.4% to 43.5% for control mothers, while informal work stays around 15% to 19% in both groups. Other labor market outcomes show similar patterns. This is a first glance showing that labor market outcomes did not change significantly because of the policy. However, this is a raw mean data comparison, and not a causal effect.

Table 2 shows the characteristics for mothers in both groups before and after the policy. We can observe that the change across years for each group is similar in the majority of the variables. For instance, years of education increase in 1.5 years for treated and control mothers. The number of children and average age of mothers is also stabble over the years. The largest difference between the two groups, is whether mothers have a partner. The percentage change of this variable is 10pp for treatment group, and 13pp for the control group.

According to a few policy briefs describing this policy, one of its main goals is to help low income families. Hence, I think it is important to analyze the data (and the outcomes) across income quintiles. Figure 6 shows the share of low, medium and high income households in each group using the quintile variable of CASEN survey [9]. We can observe that the first and last quintile

[9] Quintiles are CASEN's own variables, ranking households by autonomous income per capita, labor earnings, self-

Table 1: Summary statistics of the outcome variables, treated and control women

	Pre (1998–2006)		Post (2007–2024)		
	Treated	Control	Treated	Control	Obs.
Panel A. Children (survey): youngest 1–4 of treated women (control women's children, 6–9, are in school)					
Attends JUNJI (%)	5.9	–	14.9	–	119,028
Attends public care (%)	12.0	–	25.7	–	119,028
Attends voucher establishment (%)	4.0	–	12.4	–	119,028
Attends private establishment (%)	6.0	–	6.2	–	119,028
Attends any establishment (%)	22.8	–	45.3	–	119,028
Panel B. Women 22–45 (survey)					
Works (%)	41.7	48.3	54.3	61.3	200,761
formal: contributes to a pension system (%)	26.7	29.4	38.1	43.5	199,387
informal: does not contribute (%)	14.8	18.6	15.8	17.5	199,387
Searches for work, if not working (%)	12.9	10.7	9.9	9.1	104,855
Weekly hours, if working	40.2	39.7	37.8	38.3	87,149
Monthly labor income, thousands of CLP, if working	277	268	339	338	84,781
Reason for not looking: childcare (%)	21.2	10.8	18.3	10.5	135,725
Reason for not looking: chores (%)	16.8	17.5	14.0	13.7	135,725
Reason for not looking: no interest (%)	1.4	1.0	0.5	0.5	200,761
Panel C. The 322 cities (administrative, weighted by children 0–4; JUNJI from 2000, other providers from 2004)					
JUNJI facilities per 1,000 children 0–4	1.23		2.28		7,527
JUNJI enrollment per 100 children 0–4	6.3		13.3		7,527
Municipal enrollment per 100 children 0–4	11.7		10.2		6,762
Voucher enrollment per 100 children 0–4	9.9		15.4		6,762
Private enrollment per 100 children 0–4	2.9		3.5		6,762

Table 2: Characteristics of treated and control women

	Pre (1998–2006)		Post (2007–2024)		
	Treated	Control	Treated	Control	Obs.
Age (years)	31.6	34.5	32.0	34.8	200,761
Years of schooling	11.0	10.9	12.5	12.4	199,981
Has a partner (%)	74.7	73.2	64.9	59.7	200,761
Household head (%)	9.2	12.1	25.5	30.6	200,761
Lives in an urban area (%)	89.2	89.0	88.2	87.9	200,761
Children 0–17 in the household	2.27	2.03	2.06	1.79	200,761
Household in Q1 (%)	28.9	25.6	27.0	24.1	200,188
Household in Q5 (%)	12.1	12.7	11.5	10.9	200,188
Works (%)	41.7	48.3	54.3	61.3	200,761
Weekly hours, if working	40.2	39.7	37.8	38.3	87,149
Monthly labor income, thousands of CLP, if working	277	268	339	338	84,781

shares are similar across groups, which means there are no large income differences between the treatment and control groups. However, since we know that this policy do target low-income families, is important to take that into account. Figure 7 shows that for cities considered poor [10], their enrollment and number of facilities increase is larger than middle income cities. Since the raw data is showing a positive relation between the policy take up and income segregation, I estimate the regressions and present the results by income group.

Figure 6: Household income quintile of treated and control women

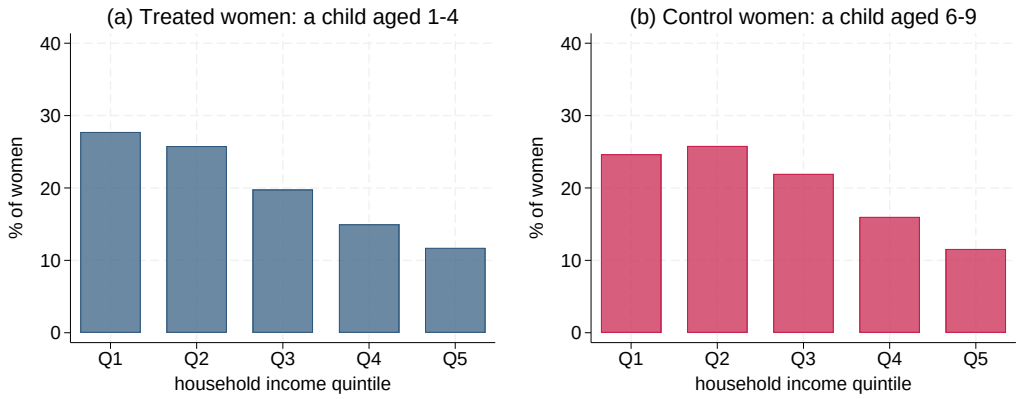
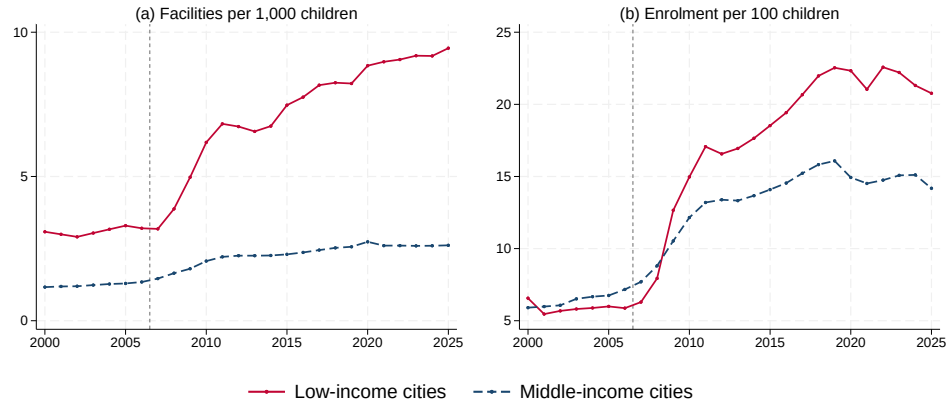


Figure 7: Facilities and enrollment by income class



In Figure 8 we can observe that richest cities do not offer nor demand public daycares as much as they do for private ones. In Panel (c) we can see that enrollment in private facilities in rich cities is approximately four times larger than private enrollment in low or medium income cities. And the opposite happens for public enrollment, showed in panel (a). One interesting finding in the raw data, is that the supply of private daycares (per 1000 children) in rich cities actually declines over

employment, capital income, pensions, and private transfers, before state monetary subsidies, with national cut-offs placing one fifth of the expanded population in each group. Q1 is the poorest fifth.

[10] Those cities that have 50% or more of households belonging to the poorest quintile (Q1).

the years. This could be due to replacement, market failures, or crowding out. However, this data in terms of private daycares is not a 100% accurate. This is due to the private childcare facilities present in the MINEDUC data are those who are officially registered in, and therefore, recognized by the Ministry of Education. But, I have confidence that I do have more than 70% of the actual number of private facilities.

Figure 8: Enrollment and facilities in private and JUNJI daycares by income class



V Empirical Strategy

I use a difference-in-difference (DID) approach to estimate the effects of a national daycare policy in Chile on mothers labor market and time-use decisions. I observe an increase of public daycare supply, lead by JUNJI (national committee of Daycares), of around 100%. Starting in 2007, new facilities opened in every city around the country. Hence, I use the age of the child to indicate eligibility and define treatment and control groups. Meaning, instead of comparing places, I compare women by the age of the children they live with, before and after the expansion, and inside the same city and year. This is the same idea behind the papers that use the age of the child as eligibility to study public childcare and preschool (Gelbach, 2002; Cascio, 2009; Berlinski and Galiani, 2007; Havnes and Mogstad, 2011).

I focus on two DID estimations. One, is the common two-by-two DID with its event study

version, and the second is a staggered DID using an imputation approach proposed by Borusyak et al. (2024). The first compares mothers of children between 1-4 years old (*eligible/treated*) with mothers of children between 6-9 years old (*noneligible/control*) before and after the policy. Secondly, I estimate a staggered version of this using the child age. This is necessary since we cannot rule out that mothers with older children in later years were not treated when their children were eligible. Hence, I construct an imputation for the control group's outcomes to have the correct counterfactual for mothers who participate of the policy using never treated and not yet treated mothers.

V.I Treatment Assignment

I use the broader household survey in Chile called CASEN from 1998 to 2024, having eleven waves (i.e., 1998, 2000, 2003, 2006, 2009, 2011, 2013, 2015, 2017, 2022, 2024). My sample consist in mothers of 22 to 45 years old with at least one children between 0-17 years old in each year-wave. The age range for mothers has two reasons. One, I wanted mothers of older children entering in the analysis are of at least 15 years old (e..g, a 22 year old mother with an 8 year old, became a mother when she was 14-15 years old). Two, the upper bound occur to minimize the possibility of studying grandmothers instead of mothers. I do not observe a direct relationship between women and children within the household, so I am assuming they are mothers (or caregivers) if they are of age and there is a child present in the household.

The treatment is defined with the age of child. A mother with a child of 1-4 years old belong to the *treatment group* ($T_i = 1$). Whereas, a mother with a child between 6-9 years old, and **no** child of 1-4 years old, belong to the *control group* ($T_i = 0$). For the two-by-two DID estimation, I do not use other mothers (e.g., of older children or newborns). However, they are useful for the staggered estimation that I will explain in Section V.IV.

There are three reasons for these definitions. First, I start the treatment group at age 1 and not at birth, because in Chile mothers have a paid maternity leave after the birth for six months, hence, children younger than 1 rarely attend a daycare. Second, I stop at age 4, because that is the last age children can attend only to a daycare. When children are 4-5 years old, they attend pre-K and Kindergarten, offered by daycares and schools. Third, the control children are between 6 and 9 years old, the strict first years of primary school, so they are close in age to the treated children, but they are in school and they **cannot use** a JUNJI daycare. This is the usual way to define a control group in this literature, as *non-eligible*: the control mothers live in the same cities and years, they face the same labor market shocks, and the only thing they cannot do is send a child to the new daycares.

V.II Difference in Difference Estimation

The main equation that I estimate is the two-by-two DID, which can be written as,

$$y_{ict} = \beta \left(T_i \times Post_t\right) + \alpha_{ct} + \gamma_{c,T_i} + \varepsilon_{ict}, \tag{12}$$

where y_{ict} is the outcome of woman i, living in city c, observed in year t. T_i indicates whether the mother is in the treatment group, and $Post_t$ is an indicator equal to one from 2007 onward. I take 2006 as the last pre-policy year, because the new facilities started opening in 2007. The coefficient of interest is β, which measures how much the outcomes of women with a child of daycare age changed because of the policy compared to what would happened in the absence of the policy. This estimator is named *average treatment effect on the treated*, and it depends on whether the parallel trend assumption is satisfied.

The fixed effects (FE) in this equation are represented by α_{ct} and γ_{c,T_i}. The first one is a set of *city by year* FE. They absorb everything that happens to all the women of a city in a given year: the local labor market, the local services, the economic cycle, and also the supply of JUNJI daycares of that city in that year. So, the comparison is always between treated and control women who live in the same city at the same moment, and any shock that hits both of them in the same way is taken out.

The second one is a set of *city by group* FE. These are the city-specific version of the treatment dummy. A standard two by two would include only T_i, which allows treated and control women to be different on average. But mothers of young children and mothers of school-age children are not different in the same way everywhere. In a city with a large informal sector or with grandmothers at home the gap between the two groups can be smaller than in a city where most women work in formal jobs. The city by group FE allow this gap to be different in each city, so the regression only uses how the gap *changes* after 2006 inside each city, and not differences in the level of the gap across cities. Since the share of treated and control women of each city is not the same in every wave of the survey, without them a change in this share could move β even if nothing happened to the gap inside any city.

I weight every regression by the city expansion weights of the survey, and standard errors are clustered at the city level (in some columns I add the woman's age, schooling and household role as controls), which is the level at which the supply of daycares and the labor market are shared (Bertrand et al., 2004).

I estimate several outcomes, and I divide them in four groups. First, *take-up*, using the survey I use a dummy variable indicating whether the child goes to a JUNJI daycare (I also check attendance to any public daycare [11]). I also use the administrative enrollment at the city level to estimate a

[11] I use this variable to relax a possible measurement error coming from parents' incorrectly answering where their children go to daycare. It is easy to mistake daycare providers among public supply.

different version of take up explained later.

Second, *labor market outcomes*, this are at the mothers' level using survey data, I have two dummy variables, one indicating whether the mother is working or not, and another to indicate whether the mother (conditional on not working) is looking for a job. I also have the log of income and weekly working hours, the latter to measure the intensive margin of working.

Third, I use variables representing mothers' time use. Specifically, I use three dummy variables that I create from the answer to the question: "Why you did not look for a job," so we can assume that these mothers are inactive. The variables indicate three main reasons (separately), one is "because of childcare constraint", "not interested in working", and "because they have to do housework." The idea for these variables is to represent the use of time for inactive mothers, and their outside options for not being working.

Fourth, *substitution* effects. Here I am concentrating on enrollment in other type of daycares (municipal, private, and voucher). I use an indicator variable for attendance to these daycares, as well as, the administrative records of enrollment.

These outcomes test the first prediction of the model (Section II). Since JUNJI is cheaper and of higher quality than a municipal daycare (equation 6), a mother whose child is in a municipal daycare moves the child to JUNJI as soon as she gets a place, while a mother who uses a private daycare moves only if her household income is below a threshold. A switch between two daycares does not change the time the mother has, while a move from home care to a daycare does (equation 11), so these outcomes tell me how much of the new JUNJI enrollment freed time for mothers and how much was a change of provider.

To see the path of the effect and to check the pre-period, I also estimate the DID event-study version,

$$y_{ict} = \sum_{\substack{s=1998 \\ s\neq 2006}}^{2024} \beta_s \left(T_i \times \mathbf{1}\{t=s\}\right) + \alpha_{ct} + \gamma_{c,T_i} + \varepsilon_{ict}, \tag{13}$$

where s runs over the survey waves and 2006 is the omitted wave. Every β_s is the difference between treated and control women in year s, relative to what that difference was in 2006. The coefficients before 2006 should be zero if the two groups were moving together before the policy, so for every outcome I report the joint test that $\beta_{1998} = \beta_{2000} = \beta_{2003} = 0$, which is the *pre-trend* test of every table.

V.III Identification Strategy Intuition

For the estimator β be a causal effect of the policy obtained using a DID I need to satisfy some assumptions. First, and usually most important, is *parallel trends*. In the context of this paper, this

assumption looks like if the expansion had not happened, the gap between women with a child of 1 to 4 and women with a child of 6 to 9 in the same city would have stayed the same after 2006 as it was before. It does not require the two groups to look the same, and they do not, since mothers of young children work less and spend more time on childcare by definition. What it requires is that the difference between them is stable (i.e., parallel) over time, apart from the policy.

Why do I think parallel trend assumption is satisfied. The two groups are mothers in the same stage of life, a few years apart, living in the same cities and observed in the same years, so most of what changes in Chile during these twenty years, the growth of the economy, the increase in women's education, the change in social norms about women's work, reaches both groups. The difference in differences estimation from equation 12 takes out everything that affects both of them in the same way. Moreover, the treatment is not chosen by the women at the moment of the survey, since the age of the child is exogenous and fixed by a birth that happened years before.

The second assumption for DID causal estimation to hold is *no anticipation*. This assumption holds for mainly two reasons. One is that for early treated mothers (i.e., mother of a child of 1-4 years old in 2007) she could not have make the decision to have her child because of the policy. This intution hold for treated mothers in early years after the policy. One could think that in later years mothers were reacting to the policy by having more children, since the cost of children decreased for them. However, I show that the policy does not have an effect on fertility, and even more, in Chile fertility rate have been decreasing since the last decades (I show this in Figure 34, in Appendix E.)

The third assumption is *no interference (SUTVA)*. This assumption consist on that the expansion affects only the women whose children are eligible, and I believe this assumption holds in this context. First, the comparison is between women of the same city and year, so any effect of the expansion on local wages or childcare prices that reaches all mothers of a city is absorbed by the city by year fixed effects. Second, mothers usually choose a daycare close to where they live, since in an emergency they rely on family and neighbors who live nearby (no cross-city spillover effects). Third, the effect of new facilities on local labor market condition if exits it cannot be too strong or significant for untreated mothers be affected. This is because the new facilities, at best could created jobs, which relative to the size of local labor markets is a small share. Fourth, private childcare does not respond to the expansion: private enrollment stays at about five children per 100, and private and voucher care in the survey does not move away from its trend. Finally, I find no channel strong enough to move the effect between eligible and ineligible mothers, beyond the exposure of control women when their own children were young, which the staggered version addresses.

V.IV Staggered Difference in Difference Estimation and Imputation

One honest caveat that the common DID approach present in this paper is the weakness of the treatment group. The weakens comes from a combination of the repeated-cross section data and child age of mothers in the control group in later years. Specifically, it could be the case that a mother in the control group in 2022 with a child of 7 years old, was treated from 2016-2019 (2017 survey wave). Difference combinations of survey year and child birth year can make this a string enough problem. Hence, I use a staggered DID method so I can use as control group never treated and not yet treated mothers (Goodman-Bacon, 2021; Callaway and Sant'Anna, 2021; Sun and Abraham, 2021; De Chaisemartin and d'Haultfoeuille, 2024; Borusyak et al., 2024).

Rationalizing this, a child observed at age a in the survey year t was born in $b = t - a$, and was between 1 and 4 years old in the years $b+1$, $b+2$, $b+3$ and $b+4$. This child gave the mother access to the expansion if at least one of those years is 2007 or later, which happens when the child was born in 2003 or later. A woman is *exposed* in year t, $E_{it} = 1$, if at least one of her children was between 1 and 4 years old in some year between 2007 and t. With this rule, every woman in every wave is in one of four groups:

- **Exposed now (N)**: she is exposed, and her youngest child is between 1 and 4 years old in year t. These are the treated women of the two by two after 2007.
- **Exposed before (B)**: she is exposed, but her youngest child is already 5 or older in year t. Most of the control women of the two by two after 2009 are in this group.
- **Never exposed, or not yet exposed (U)**: every woman before 2007, and after 2007 the women whose children are all infants (they are not 1 yet, so they are not exposed yet) or were born in 2002 or before (they were already 5 in 2007, so they were never exposed).
- **Will be exposed (L)**: women observed before 2007 whose child will be between 1 and 4 years old in 2007 or later. They are not exposed when I observe them, and I use them to test the assumption of the estimator.

Table 3 shows the rule as a calendar. Each row is the birth year of the youngest child, each column is a survey wave, and each cell gives the age of the child in that wave and the status of the mother. You can observe in the row of a child born in 2005 that the mother is a lead in 2006, when the child is 1, she is exposed now in 2009, when the child is 4, and she is exposed before from 2011 on, when the child is 6, 8 and 12. A child born in 2002 never gives access to the policy, in 2006 the child is 4, and in 2007 the child is already 5. You can also observe that before 2007 all the women are not exposed, and that after 2007 the only women who are not exposed are the mothers of infants and the mothers of children born before 2003, who become fewer every wave until they disappear in 2022.

Table 3: Exposure calendar: age of the youngest child in each wave and status of the mother

Youngest child born in	2003	2006	2009	2011	2013	2017	2024
1998	5: U	8: U	11: U	13: U	15: U	19: –	26: –
2000	3: U	6: U	9: U	11: U	13: U	17: U	24: –
2002	1: U	4: U	7: U	9: U	11: U	15: U	22: –
2003	0: L	3: L	6: *B*	8: *B*	10: *B*	14: *B*	21: –
2004	–	2: L	5: *B*	7: *B*	9: *B*	13: *B*	20: –
2005	–	1: L	4: **N**	6: *B*	8: *B*	12: *B*	19: –
2006	–	0: L	3: **N**	5: *B*	7: *B*	11: *B*	18: –
2008	–	–	1: **N**	3: **N**	5: *B*	9: *B*	16: *B*
2010	–	–	–	1: **N**	3: **N**	7: *B*	14: *B*
2011	–	–	–	0: U	2: **N**	6: *B*	13: *B*
2012	–	–	–	–	1: **N**	5: *B*	12: *B*
2014	–	–	–	–	–	3: **N**	10: *B*
2020	–	–	–	–	–	–	4: **N**
2024	–	–	–	–	–	–	0: U

Note: Each cell shows the age of the youngest child in the survey wave and the status of the mother. U: not exposed (never exposed, or not yet exposed). **N**: exposed now (youngest child 1 to 4). *B*: exposed before (youngest child 5 or older). L: will be exposed (lead). –: not born yet, or 18 or older. A woman with several children is exposed if any of them was between 1 and 4 years old after 2007; the table follows only the youngest.

In a standard staggered design, units are treated at different dates, and the estimator compares the units that are already treated with the units that are not yet treated at the same date (Callaway and Sant'Anna, 2021; Sun and Abraham, 2021). This does not work here, because the date at which a woman is first exposed is set by the birth of her child. A mother whose child was born in 2008 is first exposed in 2009, when the child is 1, and a mother whose child was born in 2010 is first exposed in 2011, also when the child is 1. Meaning, the years since exposure are always the age of the child minus one, and the women who are not yet treated at any date are the mothers of infants. A standard estimator would compare mothers of 3-year-olds with mothers of infants and call the difference an effect. But that difference is mostly how the life of a mother changes as her child grows, she comes back from the maternity leave, she goes back to work, she has more time, and this is true with or without the policy (Kleven et al., 2019). The age of the child is both the event time and something that changes the outcomes by itself, so I need an estimator that learns how outcomes change with the age of the child from women who were not exposed, and then compares.

I use the imputation estimator of Borusyak et al. (2024). The idea is simple. First, I learn from the women who were not exposed how the outcomes depend on the city and year and on the age of the youngest child. Second, I use this to predict what each exposed woman would have done without the policy. Third, I compare what her actual outcome is, with this prediction, and I average them.

To explain it in more details, in the first step, I estimate the outcomes of the women who were not exposed only ($E_{it} = 0$),

$$y_{ict} = \alpha_{ct} + \lambda_{k(i)} + \kappa_{n(i)} + \varepsilon_{ict}, \qquad \text{if } E_{it} = 0, \tag{14}$$

where $k(i)$ is the age of the youngest child of woman i, from 0 to 17, and $n(i)$ is her number of children (1, 2, 3, and 4 or more). There are three sets of FE, and each of them comes from a different group of women.

- α_{ct}, *city by year* FE. They capture what is common to all the women of city c in year t, the local labor market, the local services and the questionnaire of that year. Before 2007 they are estimated from all the women. After 2007 they are estimated only from the women who were not exposed, the mothers of infants, and the mothers of children born before 2003 (mothers with older children).

- λ_k, *age of the youngest child* FE. They capture how mothers differ by the age of their youngest child. After 2007 every woman whose youngest child is between 1 and 4 is exposed, and from 2011 on every woman whose youngest child is between 1 and 9, so for these ages the FE come almost only from 1998, 2000, 2003 and 2006. Meaning, they measure the gap between, for example, mothers of 3-year-olds and mothers of infants before the policy.

- κ_n, *number of children* FE, since a mother of three children who has a 3-year-old is not in the same situation as a mother with only one child of 3 years old.

In the second step, for every exposed woman I predict the outcome she would have had without the policy,

$$\hat{y}_{ict}(0) = \hat{\alpha}_{ct} + \hat{\lambda}_{k(i)} + \hat{\kappa}_{n(i)}, \tag{15}$$

which is what the women of her city and year who were not exposed are doing, plus the usual gap of mothers whose youngest child has the age of her child, plus the usual gap of her number of children. In the third step, her effect is the difference between what she does and this prediction, $\hat{\tau}_{it} = y_{ict} - \hat{y}_{ict}(0)$, and I average it over groups of exposed women. I report two averages: over the women whose youngest child is between 1 and 4 years old (*exposed now*), and over the women whose youngest child is between 6 and 9 (*exposed before*), the same ages as the control children of the two by two. I also report the average for each age of the youngest child. The first one is the effect while the child can attend a daycare, and the second one is whether that effect lasts after the child goes to school, which is the part that the two by two subtracts.

These three steps are the same as estimating one regression on the whole sample,

$$y_{ict} = \alpha_{ct} + \lambda_{k(i)} + \kappa_{n(i)} + \sum_{(t',a)} \tau_{t',a} \mathbf{1}\{E_{it} = 1,\ t = t',\ k(i) = a\} + \theta_{03} L_{it}^{03} + \theta_{06} L_{it}^{06} + \varepsilon_{ict}, \quad (16)$$

where each group of exposed women observed in the same year t' and with a youngest child of the same age a has its own dummy. Since every exposed woman is inside one of these dummies, the exposed women do not help to estimate α, λ or κ, those are estimated only with the women who were not exposed, and each $\tau_{t',a}$ is the average effect of the exposed women in that cell. The effects I report are the averages of the $\tau_{t',a}$ weighted by the number of women in each cell, with their standard errors from the same regression. L_{it}^{03} and L_{it}^{06} are indicators for the women who will be exposed, observed in 2003 and 2006, and I explain them below. As in the difference in differences, I weight by the city expansion weights of the survey and cluster the standard errors at the city level.

The identification assumption for the imputation method is the following. Without the policy, the gap between mothers whose youngest child has a given age and the women of the same city and year who were not exposed would have stayed as it was before 2007. In the example, without the expansion, mothers of 3-year-olds in 2011 would differ from mothers of infants and of older children in their city in the same way as mothers of 3-year-olds did before 2007. This is a parallel trends assumption, like the one of the difference in differences, but the comparison is different. The two by two compares mothers of young children with mothers of school-age children who were exposed before. The imputation compares them with women who were never exposed, and it uses the time before the policy to learn how mothers differ by the age of their children. So, the effect of having been exposed before is not subtracted, it is estimated.

The lead test. The way I check this assumption is with the women who will be exposed. For instance, assume a mother is observed in 2003 with a newborn, and another women is observed in 2006 with a child of 2 years old, are not exposed when I observe them, but their children will be between 1 and 4 years old after 2007. If the women whose children will be exposed were already different before the policy, more than what the age of the child explains, the coefficients θ_{03} and θ_{06} would not be zero. For every outcome I report the joint test that $\theta_{03} = \theta_{06} = 0$. There is no lead in 1998 or 2000, because a child observed in 1998 (2000) was born in 1998 (2000) or before and was already 5 years old in 2003 (2005), so no woman observed in 1998 (2000) lives with a child who will be between 1 and 4 years old after 2007; the 1998 (2000) women help to estimate the age FE and the city by year FE of that year. This test plays the same role as the pre-trend test of the event study (Borusyak et al., 2024). It only exists for outcomes that are measured before 2006, so for childcare and chores as reasons for not looking for work, which start in 2006, there is no lead test, and their age FE come only from the 2006 wave.

V.V Measuring Take Up

Mothers who have a child of daycare age and send their child to a JUNJI daycare (after the policy) are known in the DID literature as *compliers*. However, in this context the DID estimation (also the staggered version presented above) is more complex for the so called "first stage" or "take up." Even though I observe compliers mothers, their counterfactual, or in this case, the control group are also compliers of their assignment by definition. The control group is mothers of older children who are not allowed to attend JUNJI daycares. Hence, the DID estimation for this outcome is not straightforward.

I measured the take up at the child level, using every child between 1 and 4 years old (treated) and between 5 and 9 years old (control), whose outcome is school attendance. The outcomes are whether the child attends a JUNJI daycare, a public establishment (JUNJI, INTEGRA or a municipal one), or any educational establishment. Equation (12) can be estimated for these outcomes, but it is important to be careful with what it measures. The control children cannot attend JUNJI, so their JUNJI attendance is zero in every year. This is what makes them a good control for the mothers' outcomes, and it is the definition of non-eligible. However, a group whose outcome is fixed (but not because they are necessary compliers) does not tell me anything about how the attendance of the treated children would have moved without the policy. Meaning, for the take up results, β is just the change in the treated children's own attendance after 2006, and the assumption behind it is that their attendance would have stayed flat. If JUNJI attendance was already increasing before 2006, which the pre-trend test can show, the two by two counts that increase as part of the policy. For this reason, I measure the take up results in two additional ways that do not need a flat counterfactual.

The first one is a *trend break* in the survey. Using only the linked child of the treated women, I estimate

$$y_{ict} = \alpha_c + b\,(t-2006) + \delta\, Post_t + \sigma\,(t-2006) \times Post_t + \varepsilon_{ict}, \tag{17}$$

where α_c are city FE, b is the trend before the policy, δ is the jump in 2007, and σ is the change in the slope after 2006. Here the counterfactual is not zero, it is that the attendance would have continued along the line it was following before 2006. I report the joint test that $\delta = \sigma = 0$, and how far the attendance is from its old line by 2024, which is $\delta + 18\sigma$. I also estimate the detrended event study, where the jump and the change in slope are replaced by one coefficient for each wave after 2006. I estimate this for JUNJI, but also for INTEGRA, municipal establishments, and private and voucher establishments, because a new JUNJI place can be taken by a child who was already in another establishment, and then the expansion moves children between providers instead of adding new ones. This latter results are presented in the substitution subsection in Results, showing how

mothers (already working) changed providers instead of removing her childcare constraint.

The second one uses the administrative records of JUNJI, which list every facility with its program and its enrollment every year. I estimate equation (17) on the JUNJI enrollment per 100 children between 0 and 4 years old at the city level, for the cities of my sample that have JUNJI records every year between 1998 and 2024, weighted by the child population. The child population starts in 2000, so for 1998 and 1999 I divide by the population of 2000. Moreover, the administrative data let me separate the facilities that opened in 2007 or later from the ones that already existed. The enrollment in the new facilities is zero before 2007 by construction, and it would have stayed at zero if the expansion had not happened, so for this outcome the counterfactual is exactly zero and does not need any assumption about trends. In this case the logic of the non-eligible control holds exactly, since no child could attend a facility that did not exist yet. I also estimate the trend break for the old facilities alone, because if the old facilities stop growing when the new ones open, part of the new enrollment is a replacement and not an increase.

For the staggered and imputation version the logic is different. The same logic cannot be used for the attendance of the children. After 2007 every child between 1 and 4 years old is exposed, and the children who are not exposed are the infants, who rarely attend a daycare and who could be partly reached by the expansion of nursery places, and the children of 9 years old and older, who cannot attend JUNJI at all. So, there is no child who was not exposed whose JUNJI attendance can tell me what the attendance of an exposed child would have been. For this reason, the first stage of the staggered version is the enrollment in the new facilities of the administrative data, which has a counterfactual of exactly zero, as I explained before. I also report the imputation at the child level, together with its lead test, as a complement.

VI Results

I study a large public daycare expansion in Chile on mother's labor market outcomes and the mechanisms behind it. In 2006, the government launched a public program concentrated on help families and children in their life. One of the layers of the program was increase the public supply of JUNJI daycares. The expansion was large and significant, creating about 1,500 new facilities in the cities of my sample (almost doubling the supply). One should expect that this impacted mothers on their capacity to participate in the labor market. Nevertheless, I find null effects on employment and job search.

To understand this null response, I estimate two possible mechanisms. First, part of the increase in enrollment on these new slots come from children already enrolled in other daycares. I show a decrease in municipal daycare enrollment caused by the policy, indicating a *substitution effect* on mothers daycare preferences. I find that women change providers among public supply. There is significant decrease in municipal daycare enrollment of 6.2 children per 100 below its pre-

2006 trend. Justifying the null response on employment, since those mothers who changed were probably already working.

Second, I can observe the reason mothers give to explain why they are inactive. I show that using "childcare" decreases because of the policy in 1.9 percentage points among all women, but "household chores" increases in 2.1 percentage points among inactive mothers. Demonstrating that mothers stay out of the labor force not only because of childcare barriers. Hence, if childcare policies are done to help mothers join the labor force, we need to understand all their barriers.

The main results are estimated using equation 12. The outcomes are presented in chronological order. First, the take-up of the policy, showing JUNJI attendance as the principal outcome. Second, labor market outcomes, where I have an indicator of whether the mother is working, and another indicator for whether the *non-working* mother looked for a job. I also have the weekly hours and salary for working mothers. Third, I present the substitution effects across daycare providers. Fourth, I have three variables to represent mothers' time allocation preferences. These results are presented in tables, showing equation 12 results, and their respective event studies presented in equation 13.

In Chile, employment can be broadly classified into two categories: *formal* and *informal* employment. I use pension contributions as a proxy for employment formality, classifying mothers according to their responses to the CASEN question on whether they currently contribute to a pension system. This approach is consistent with commonly used measures of labor market informality in Chile (OECD, 2022). Distinguishing between formal and informal employment is particularly relevant in the context of public daycare expansion, as access to childcare may affect not only mothers' decisions to participate in the labor market but also the types of jobs they obtain. Formal employment generally provides greater access to social protection and employment benefits, while informal employment may offer greater flexibility but less job security. These differences may influence how mothers respond to increased childcare availability, particularly given the challenges of reconciling childcare responsibilities with fixed work schedules.

I finish the main analysis by studying the effects on the outcomes by income quintile. Due to the policy's target are low-income families, it is important to test whether these results are concentrated in one income group. Take-up effects are concentrated in the middle income group, and the substitution effect in the low and middle income groups. While labor market results stay null in the income analysis, and the change in time use is concentrated in the poorest households. I also present robustness checks using the staggered imputation DID estimation (Borusyak et al., 2024), where the time use results hold and get stronger, and the increase in informal work is concentrated while the child is in daycare age.

VI.I Take - Up

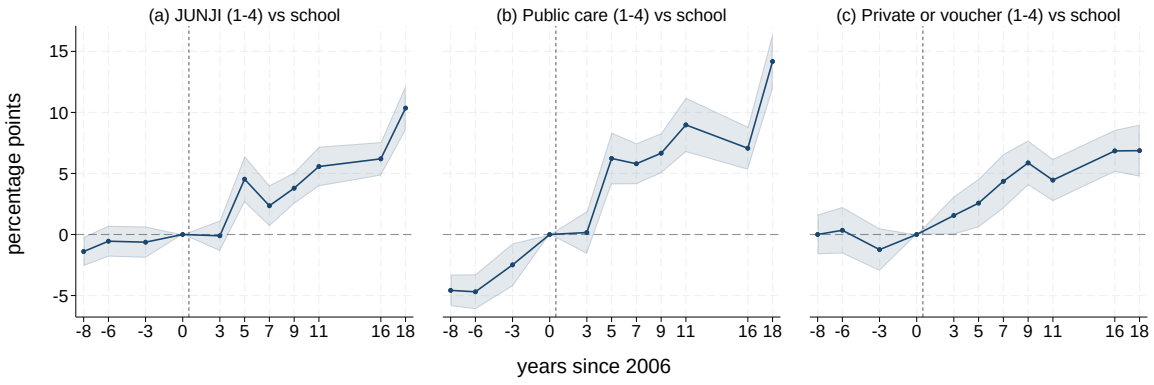
The direct effect of a childcare policy is increasing children enrollment. I estimate the effects of the policy on attendance in four different group of daycares. Table 4 shows the DID results of attendance in JUNJI, public, private or any daycare. In particular, JUNJI attendance of young children rises 5.2 percentage points relative to the school attendance of older children, more than half of the 9.2 percent who attended in 2006 (Table 4). Figure 9 shows that the gap opens gradually after the policy, and that by 2024 it is 10.4 points. In the administrative records, the facilities opened from 2007 enroll 9.8 children per 100 by 2024 (Table 7).

Table 4: Take-up at the child level: children 1–4 against children 5–9

	Mean 2006 (children 1–4)	Two by two	Event study, mean post-2006	Pre-trend p
Attends JUNJI	9.2	5.2***	4.7***	0.15
		(0.4)	(0.6)	
Attends public care	18.4	9.8***	7.0***	<0.01
		(0.5)	(0.7)	
Attends private or voucher	12.5	4.8***	4.6***	0.25
		(0.5)	(0.7)	
Attends any establishment	31.7	19.0***	13.9***	<0.01
		(0.6)	(0.8)	
Observations		320,321	320,321	
Children	136,347 aged 1–4 (treated), 183,977 aged 5–9 (comparison)			

Notes: Take-up at the child level: every child aged 1–4 (treated) and 5–9 (comparison) in the 322 cities, CASEN 1998–2024, whoever the child lives with. The outcome of a treated child is attendance (pp) at JUNJI, at public care, at a private or voucher establishment, or at any establishment; the outcome of a comparison child is attendance at school (any establishment). Coefficient on treated × post-2006 with city × year and city × group fixed effects, survey weights. Event study: mean of the 2009–2024 coefficients, 2006 omitted. Pre-trend p: joint test of 1998, 2000 and 2003. The women's controls (age, household role, schooling) do not apply at the child level. Standard errors clustered by city in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 9: Take-up at the child level: children 1–4 against children 5–9, JUNJI (a), public care (b) and private or voucher (c) against school, relative to 2006



VI.II Labor Market

Mothers' labor force response to the policy was mainly null. In terms of participation, the two indicators of working and looking for a job, are small and insignificant. Employment of treated women changes by -0.5 percentage points (standard error 0.7) over a rate of 44.7%, and searching for a job decreases in 0.8 percentage points (SE 0.7) over a rate of 11.2% (Table 5). For mothers who are currently working the policy decreased their amount of hours worked per week and their monthly salary. Specifically, weekly hours decreased by 4.9 percent (standard error 1.4), about two hours a week, and their earnings by a similar amount, which suggests that their hourly pay did not change much. Hence, overall there is not evidence for a significant effect of the policy on mothers employment.

Figure 10 shows the results in an event study format. Parallel trend assumption is violated for employment, and there is no a clear effect on the outcomes either. Figure 11 shows that employment of treated women grows smoothly through these years, while employment of control women jumps in 2006 and falls back in 2009 . The gap between the two groups is between 4 and 8 points in every wave except 2006, when it is 11. A reference year in which the controls are unusually high makes any gain of the treated look larger, not smaller, so it cannot be hiding an employment response. Using 1998–2003 as the reference instead, employment falls 2.0 percentage points (standard error 0.8), and 1.1 points, not significant, with all the controls (Table 19).

Figure 10: Labor market, event study: treated minus control women relative to 2006



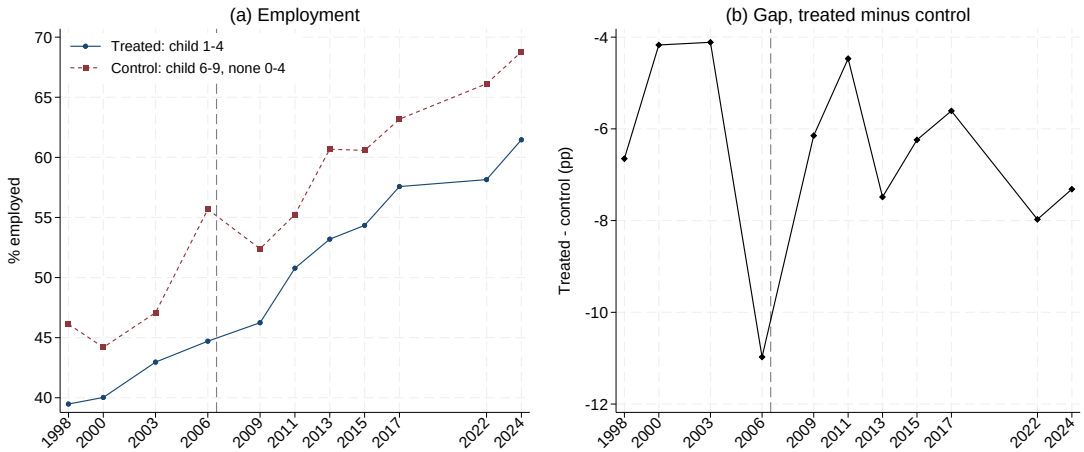
In Chile, there are two main categories of jobs, *formal* and *informal* jobs. I use a proxy variable to classify mothers in formal or informal jobs. I focus on the CASEN question that indicates

Table 5: Labor market, two by two difference in differences

	Mean 2006	(1)	(2)	(3)	(4)	(5)
Works	44.7	-0.5	-0.6	0.1	-0.5	0.2
		(0.7)	(0.7)	(0.7)	(0.7)	(0.7)
		[<0.01]	[<0.01]	[<0.01]	[<0.01]	[<0.01]
Searches for work	11.2	-0.8	-0.7	-0.7	-0.9	-0.7
		(0.7)	(0.7)	(0.7)	(0.7)	(0.7)
		[0.33]	[0.27]	[0.30]	[0.34]	[0.28]
Log weekly hours	38.3	-0.049***	-0.049***	-0.045***	-0.048***	-0.044***
		(0.014)	(0.014)	(0.014)	(0.013)	(0.014)
		[0.05]	[0.05]	[0.05]	[0.08]	[0.07]
Log monthly labor income	276,790	-0.037**	-0.038**	-0.040***	-0.033**	-0.036**
		(0.015)	(0.015)	(0.015)	(0.014)	(0.014)
		[0.83]	[0.84]	[0.81]	[0.96]	[0.94]
Age (linear)		No	Yes	No	No	Yes
Household role (indicators)		No	No	Yes	No	Yes
Years of schooling (linear)		No	No	No	Yes	Yes
Observations		200,759	200,759	200,759	199,979	199,979

Notes: Two by two difference in differences, women 22–45 in 322 cities, CASEN 1998–2024. Treated: lives with a child aged 1–4; control: no child aged 0–4 and a child aged 6–9. Works and searches for work in pp; hours and income in logs, for women who work. Observations: the regression of the first row; in column (1), Searches for work, 104,840; Log weekly hours, 87,127; Log monthly labor income, 84,756. All columns include city × year and city × group fixed effects; columns (2) to (5) add, as linear controls, the woman's age and her years of schooling (top-coded at 20), and indicators for her role in the household (head, spouse, other). Women with missing schooling drop from columns (4) and (5). Pre-trend p in brackets: joint test that the event-study coefficients of 1998, 2000 and 2003 are zero (– when the outcome starts in 2006). Mean 2006: mean among treated women in 2006, weighted (for log outcomes, the mean of the level). Standard errors clustered by city in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 11: Employment by year: treated and control women (a), and the gap between them (b)



whether the person contributes to their pension fund. This accurately categorize people regarding their job status (OECD, 2022). Table 6 shows that formal work falls 2.7 percentage points (standard error 0.8), from 29.3% in 2006, and informal work rises 2.1 points (standard error 0.6), from 14.9, so the two movements roughly cancel and leave employment where it was. However, from a public policy perspective, this is an important result, due to the lack of social security that informal jobs have.

Figure 12 shows these results year by year. For formal work, panel (a), the coefficients before the policy are between 2.5 and 3.8 points above 2006, and after the policy they move slightly below zero, between -0.7 and -1.4 points from 2013 on, so the decline grows over time. Informal work, panel (b), does the opposite: it is between 3.5 and 5.6 points above its 2006 level in every wave after the policy. As with employment, the coefficients before 2006 sit above zero for both outcomes, because 2006 was an unusual year for the control mothers, and the pre-trend tests are at the limit ($p = 0.05$). The informal share of working mothers, panel (c), which does not depend on how many mothers work, has a pre-trend test that passes ($p = 0.44$) and is between 3.3 and 6.6 points higher in every wave after 2006, so the composition of mothers' jobs moves toward informal work after the policy.

Table 6: Formal and informal work

		(1)	(2)	(3)	(4)	(5)
Works, formal (contributes)	mean 2006: 29.3	-2.7***	-2.8***	-2.2***	-2.7***	-2.2***
pre-trend $p = 0.05$		(0.8)	(0.8)	(0.8)	(0.8)	(0.8)
Works, informal (does not contribute)	mean 2006: 14.9	2.1***	2.1***	2.3***	2.1***	2.3***
pre-trend $p = 0.05$		(0.6)	(0.6)	(0.6)	(0.6)	(0.6)
Informal share among workers	mean 2006: 33.6	3.8***	3.8***	3.6***	3.6***	3.4***
pre-trend $p = 0.44$		(1.1)	(1.1)	(1.1)	(1.1)	(1.1)
Age (linear)		No	Yes	No	No	Yes
Household role (indicators)		No	No	Yes	No	Yes
Years of schooling (linear)		No	No	No	Yes	Yes
Observations		199,385	199,385	199,385	198,620	198,620

Notes: Women 22–45, 322 cities, CASEN 1998–2024. A working woman is formal if she contributes to the pension system and informal if she does not; formal and informal work are in pp of women (they add up to employment), the informal share in pp of working women. Two by two difference in differences, treated: lives with a child aged 1–4, control: no child aged 0–4 and a child aged 6–9; city $\times$ year and city $\times$ group fixed effects; pre-trend p: joint test of the 1998, 2000 and 2003 event-study coefficients, column (1). Columns (2) to (5): add, as linear controls, the woman's age and her years of schooling (top-coded at 20), and indicators for her role in the household (head, spouse, other). Mean 2006: treated women in 2006. Observations: formal and informal work, all women; the informal share uses working women (92,295 in column (1)). Standard errors clustered by city in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 12: Formal and informal work, event study relative to 2006: works formal (a), works informal (b), informal share of workers (c)



VI.III Understanding the Null Effect

Following the model presented in section II, we can designate this null response of mothers toward the policy to childcare preferences and time allocation decisions. I present these mechanisms separated in two sections. I stat with the substitution between public providers and then I move to time allocation decisions. Both of them tell a story of why mothers could use the policy but not participate in the labor force.

VI.III.1 Substitution Effects

The policy is accomplished in its full if two things happen. First, it is able to move a child from home, where the mmother is the caregiver, to a JUNJI facility. Second, the mother that was freed from childcare starts working. A JUNJI place adds care only if the child who takes it would otherwise have been at home; if the child would have been in another establishment, the place changes the provider and leaves the mother's day as it was. To see which happened, I follow each provider against its own path before 2006, in the survey and in the administrative records (Table 7). The survey cannot measure informal care, so what I can measure is the movement between establishments.

The answer comes from municipal care. In the survey, the municipal attendance of treated children follows its old path until 2013, then falls below it, and by 2024 it is 6.0 points below (Figure 13). Private and voucher care do not fall. The administrative records show the same

movement with more precision and with the timing of the policy (Figure 14). In 2007, the year the new facilities open, JUNJI enrollment jumps and municipal enrollment falls, and by 2024 JUNJI is 5.6 children per 100 above its old path while municipal schools are 6.2 below theirs. Voucher and private schools, by contrast, did not lose children. Even inside JUNJI, the facilities that existed before 2007 grew less than they would have, so part of the growth of the new facilities replaced growth of the old ones.

This resolves the puzzle of take-up. The new JUNJI places filled up, but largely with children who would otherwise have been in a municipal establishment, which is also public and in most cases run by the same municipality. For many families the expansion changed the name of the place their child attends more than whether the child is in care, and it gave their mothers no new time to work. This is the crowd-out that Havnes and Mogstad (2011) find for Norway, here between two public providers rather than from informal to formal care.

Figure 13: Survey: attendance of treated children relative to the pre-2006 trend, public care (a), INTEGRA (b), municipal (c), private or voucher (d)

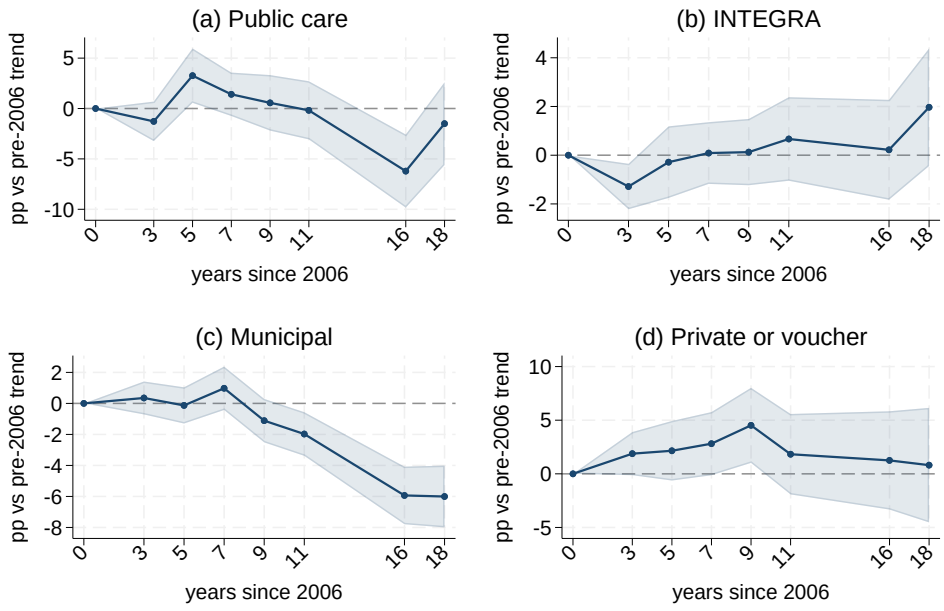
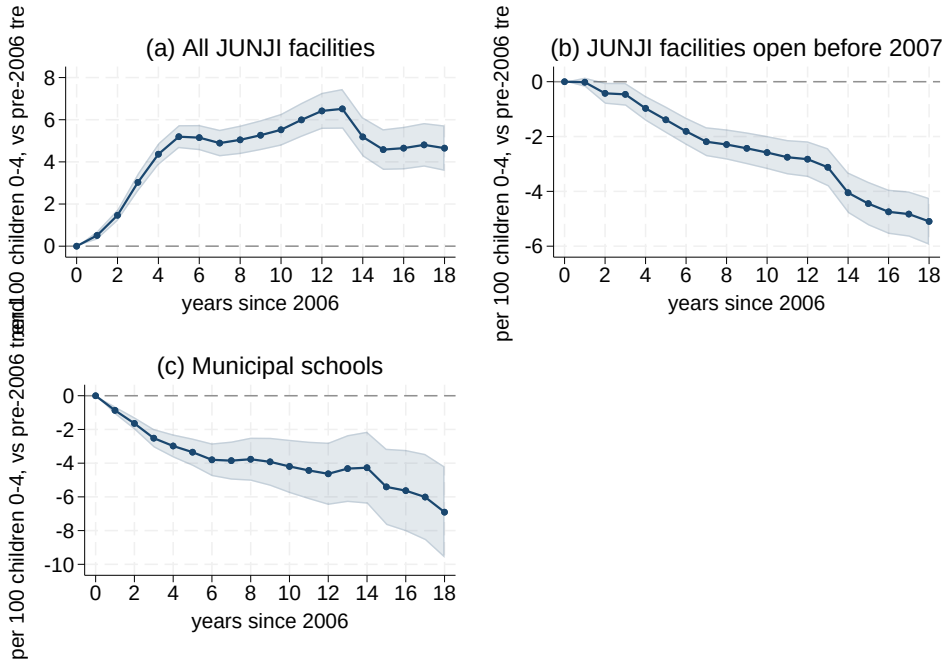


Table 7: Substitution: attendance and enrollment relative to the pre-2006 trend

	Pre-2006	Jump	Change in	Distance from old trend, 2024	
	yearly slope	in 2007	yearly slope	Baseline	All controls
Panel A: Survey, attendance (pp) of the youngest child 1–4 of treated women					
JUNJI	0.57***	1.3*	-0.01	1.1	0.2
	(0.06)	(0.8)	(0.08)	(1.5)	(1.6)
Public care	1.03***	2.9***	-0.35***	-3.4*	-4.6**
	(0.08)	(0.9)	(0.10)	(2.0)	(2.0)
INTEGRA	0.21***	-1.2***	0.15**	1.4	1.0
	(0.05)	(0.4)	(0.06)	(1.1)	(1.1)
Municipal	0.25***	2.9***	-0.49***	-6.0***	-5.8***
	(0.05)	(0.4)	(0.05)	(1.0)	(1.0)
Private or voucher	0.45***	3.2***	-0.10	1.4	-0.3
	(0.11)	(0.8)	(0.13)	(2.6)	(2.4)
Observations	119,028	119,028	119,028	119,028	118,563
Panel B: Administrative records, enrollment per 100 children 0–4					
JUNJI, all facilities	0.200***	3.3***	0.13***	5.6***	–
	(0.019)	(0.2)	(0.03)	(0.6)	
JUNJI, open before 2007	0.198***	0.23	-0.30***	-5.1***	–
	(0.020)	(0.18)	(0.02)	(0.5)	
Municipal schools	0.26***	-1.56***	-0.26***	-6.2***	–
	(0.07)	(0.13)	(0.08)	(1.4)	
Voucher schools	1.32***	3.2***	-1.19***	-18.1***	–
	(0.13)	(0.2)	(0.14)	(2.4)	
Private schools	0.02	-0.10	0.06	0.9	–
	(0.06)	(0.13)	(0.07)	(1.2)	
Panel C: Administrative records, levels per 100 children 0–4					
	2006	2024	Cities		
JUNJI, all facilities (national)	7.1	14.7	–		
JUNJI, open before 2007 (national)	7.1	5.0	–		
Municipal schools	12.0	9.4	312		
Voucher schools	11.7	14.6	227		
Private schools	4.7	5.5	72		

Notes: Trend breaks: outcome on a linear trend in years since 2006, a post-2006 indicator (jump) and their interaction (change in slope); distance in 2024 = jump + 18 $\times$ change in slope. City fixed effects. Panel A: CASEN 1998–2024, 322 cities, survey weights; all controls adds the mother's age and years of schooling (linear) and household-role indicators. Panels B and C: balanced city panels weighted by the child population. JUNJI from JUNJI records 1998–2024 (251 cities; 1998–99 use the 2000 population). Municipal (municipal and SLEP), voucher and private school enrollment from the Ministry of Education register, 2004–2024, cities that report the provider every year; it is mostly pre-kinder and kinder. Their pre-2006 trends rest on 2004–2006 only: the voucher distance extrapolates a steep three-year trend and should be read with Panel C. Private is reported by few cities. No individual controls apply to city-level data. Observations: Panel A, children aged 1–4 of treated women; Panels B and C, one observation per city and year. Standard errors clustered by city in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 14: Administrative records: enrollment relative to the pre-2006 trend, all JUNJI facilities (a), JUNJI facilities open before 2007 (b), municipal schools (c)



VI.III.2 Time Use Effects

If childcare were what keeps mothers of young children at home, a daycare place should make them less likely to say so. The survey asks women who neither work nor look for work why they do not look, which lets me see this directly. I measure each reason as a share of all women and of these inactive women, the population the question is asked to. The answers for childcare and chores can only be compared from 2006 on.

The two by two points in the direction the model predicts (Table 8). As a share of all women, mothers of young children become 1.9 percentage points less likely to say that childcare is why they do not look for work, about 9 percent of the 21.2 percent who cited it in 2006, and the estimate is significant in every column, between 1.7 and 2.1 points. Among inactive women, the share who cite household chores rises in every specification, by 2.1 to 2.4 points over a base of 33.8 percent, and the share who say they are not interested in working does not rise. Mothers who stay out of the labor force, in other words, move away from childcare as their reason and toward household work, and they do not reinterpret their situation as a lack of interest in working.

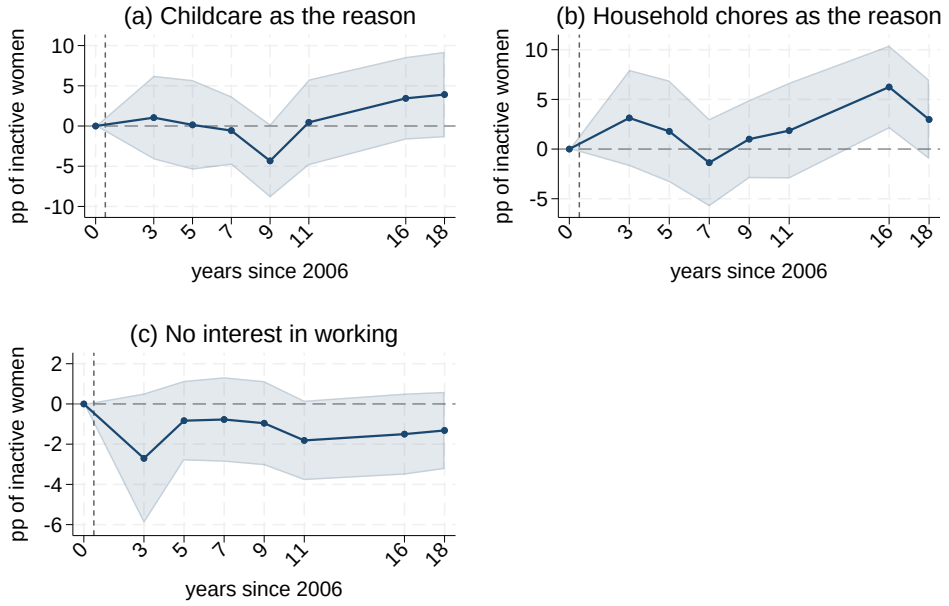
The two by two is likely to understate these movements. Its control mothers, whose children are 6 to 9, were themselves exposed to the expansion a few years earlier, so any lasting change in the reasons they give is subtracted from the estimate. The staggered comparison of Section VI.V, which compares exposed mothers with mothers who were never exposed, confirms the same pattern with larger and more precise effects: inactive mothers exposed to the policy cite childcare about 3 points less and household chores 4.5 points more.

Table 8: Time use: reason for not looking for work, two by two difference in differences

	Mean 2006	(1)	(2)	(3)	(4)	(5)
Panel A: Share of inactive women (pp)						
Childcare (2006 on)	42.7	0.4	0.4	0.5	0.4	0.4
pre-trend p = –		(2.1)	(2.1)	(2.0)	(2.1)	(2.0)
Household chores (2006 on)	33.8	2.1	2.4	2.2	2.1	2.3
pre-trend p = –		(1.7)	(1.6)	(1.6)	(1.6)	(1.6)
No interest in working (2006 on)	5.7	-1.4	-1.4	-1.4	-1.5	-1.5
pre-trend p = –		(1.1)	(1.1)	(1.1)	(1.1)	(1.1)
Panel B: Share of all women (pp), for comparison						
Childcare (2006 on)	21.2	-1.9*	-1.9*	-2.1**	-1.7*	-2.0**
pre-trend p = –		(1.0)	(1.0)	(1.0)	(0.9)	(0.9)
Household chores (2006 on)	16.8	0.4	0.4	0.1	0.6	0.3
pre-trend p = –		(0.8)	(0.8)	(0.8)	(0.7)	(0.8)
No interest in working (2006 on)	2.8	-1.1**	-1.1**	-1.1**	-1.1**	-1.1**
pre-trend p = –		(0.4)	(0.4)	(0.4)	(0.5)	(0.5)
No interest in working	2.8	-0.51***	-0.51***	-0.51***	-0.52***	-0.53***
pre-trend p = 0.05		(0.16)	(0.16)	(0.16)	(0.16)	(0.16)
No interest, without 2006 and 2009		-0.24				
pre-trend p = 0.09		(0.17)				
Age (linear)		No	Yes	No	No	Yes
Household role (indicators)		No	No	Yes	No	Yes
Years of schooling (linear)		No	No	No	Yes	Yes
Observations (Panel A, inactive women)		57,786	57,786	57,786	57,688	57,688
Observations (Panel B, all women)		135,723	135,723	135,723	135,433	135,433

Notes: Two by two difference in differences, women 22–45 in 322 cities, CASEN 2006–2024 (Panel A and the 2006-on rows of Panel B) or 1998–2024. Treated: lives with a child aged 1–4; control: no child aged 0–4 and a child aged 6–9. Outcome: the reason the woman gives for not looking for work. Inactive women: neither employed nor searching for work, the women who are asked the reason; reasons in pp of inactive women, 2006 on (the first wave in which childcare and chores are separable). There is no pre-trend or lead test for outcomes that start in 2006. Panel B divides by all women, so it also moves with employment; in 2006 employment of control women is unusually high. The last row drops the 2006 and 2009 waves, whose 21-answer questionnaire doubles the share of no interest for every woman, and uses 2003 as the reference wave (baseline only). All columns include city $\times$ year and city $\times$ group fixed effects; columns (2) to (5) add, as linear controls, the woman's age and her years of schooling (top-coded at 20), and indicators for her role in the household (head, spouse, other). Women with missing schooling drop from columns (4) and (5). Pre-trend p: joint test that the event-study coefficients before the reference wave are zero (– when the outcome starts in 2006). Mean 2006: treated women in 2006, weighted. Standard errors clustered by city in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 15: Time use, event study: reason for not looking for work, share of inactive women, treated minus control relative to 2006: childcare (a), household chores (b), no interest in working (c)



VI.IV Analysis by Income Group

The policy declares that it gives priority to low-income families for daycare access. Hence, I present results for the main outcomes by income groups. I categorize three groups, low-income (first household income quintile, Q1), middle income (second to fourth quintiles, Q2-Q4), and high-income group (fifth income quintile, Q5). Further, there is a clear correlation between people income class and their demographic characteristics. Usually, lower-income individuals have less education, more children, and lower employment rates than higher-income individuals, and these gaps are larger for women (McLanahan, 2004; Goldin, 2006; Gasparini and Marchionni, 2015).

Take Up. The JUNJI attendance of young children rises 6.6 percentage points in households of Q2 to Q4, against 2.6 points in Q1 and in Q5 (Table 9). In the middle-income group the years before 2006 are flat, showing that the pre-trend test passes (Figure 16). For the other two groups the effects are smaller, 2.6 points each, and in Q1 the pre-trend test also passes. Hence, even though the policy targets low-income families, the one who uses it are not them. This could be related to the value people give to a daycare institution compared to care at home.

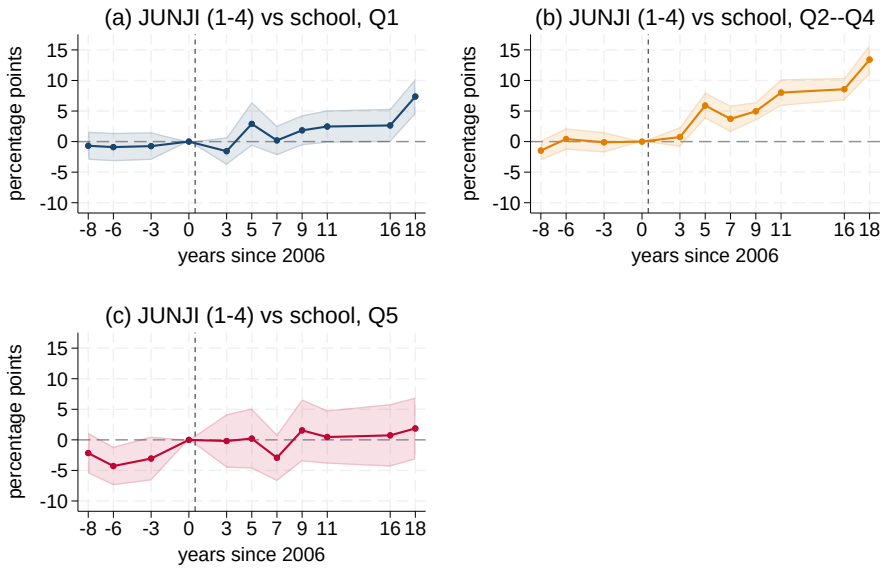
Labor Market. For labor market outcomes, Table 10 shows results of every outcome for the three income groups. Splitting the sample by household income does not reveal an employment response (Table 10). Employment of treated mothers changes by -0.7 percentage points in Q1 (standard error 1.1), -1.0 in Q2–Q4 (standard error 0.9) and 3.5 in Q5 (standard error 2.0), and

Table 9: Take-up at the child level by household income: children 1–4 against children 5–9

	Household income				
	Q1	Q2–Q4	Q5	p Q2–4	p Q5
Attends JUNJI	2.6***	6.6***	2.6**	<0.01	0.97
	(0.5)	(0.5)	(1.1)		
	[0.88]	[0.20]	[0.06]		
mean 2006	10.6	9.1	5.8		
Attends public care	8.0***	11.3***	5.0***	<0.01	0.05
	(0.6)	(0.6)	(1.4)		
	[<0.01]	[<0.01]	[<0.01]		
mean 2006	21.0	18.7	10.1		
Attends private or voucher	2.7***	5.7***	8.1***	<0.01	<0.01
	(0.6)	(0.6)	(1.6)		
	[0.15]	[0.81]	[0.60]		
mean 2006	6.6	11.4	33.4		
Attends any establishment	17.3***	20.7***	15.2***	<0.01	0.26
	(0.7)	(0.7)	(2.0)		
	[<0.01]	[<0.01]	[0.28]		
mean 2006	28.1	31.0	44.3		
Observations	116,994	179,111	23,603		

Notes: Take-up at the child level by household income: every child aged 1–4 (treated) and 5–9 (comparison) in the 322 cities, CASEN 1998–2024, split by CASEN's national income quintile of the household. The outcome of a treated child is attendance (pp) at JUNJI, at public care, at a private or voucher establishment, or at any establishment; the outcome of a comparison child is attendance at school. Coefficient on treated × post-2006 with city × year and city × group fixed effects, survey weights, each income group estimated separately. Pre-trend p in brackets: joint test of 1998, 2000 and 2003. p Q2–4 and p Q5: difference with Q1 from one regression with every term and fixed effect interacted with the group. Mean 2006: children aged 1–4 in 2006. Standard errors clustered by city in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 16: JUNJI attendance of children 1–4 against school attendance of children 5–9, event study by household income: Q1 (a), Q2–Q4 (b), Q5 (c)



adding the mothers' controls leaves the three estimates close to zero or within their confidence intervals. This holds even in the poorest households, where only one in four treated mothers worked in 2006, and where the room for the policy to bring mothers into work was largest. Searching for a job does not respond in any group either, so the expansion did not bring poorer mothers into the labor force as job seekers. The only labor-market adjustment that the income split sharpens is on the intensive margin, and it comes from the middle of the distribution: in Q2–Q4, mothers who work reduce their weekly hours by 5.4 percent and their earnings by a similar amount, while hours and earnings do not move in Q1 or Q5.

Figure 17 shows that the absence of a response is not an artifact of averaging over years, showing the event study version. In the three income groups the event-study coefficients after 2007 stay at about the level they had before 2006, and none of them shows a break when the new daycares open. As in the pooled sample, every coefficient before 2006 sits above zero because of the unusual 2006 reference year, when employment of control mothers spiked, and this is why the pre-trend test rejects in Q1 and Q2–Q4. Measured against the years 1998–2003, employment of treated mothers after the policy is, if anything, about two points lower in Q1 and Q2–Q4, and about one point higher in Q5, where the estimates are imprecise and the coefficients move around a stable level throughout the period. The positive point estimate for Q5 therefore reflects a level that was already there before the policy rather than a response to it. In no part of the income distribution, then, did the expansion of public childcare move mothers into work.

Table 11 shows that the change in the type of work is not specific to one part of the income distribution, although its form follows the jobs available in each. In the middle of the distribution, where one in three treated mothers worked in a formal job in 2006, formal work falls 3.6 percentage points (standard error 1.0) and informal work rises 2.4 points (standard error 0.7), so most of the decline in formal jobs is matched by informal ones and employment barely moves. In the poorest households, where formal work was rare (11 percent of treated mothers in 2006), formal work falls 1.9 points and informal work rises 1.2, and the change is clearest in the composition of jobs. The informal share of working mothers rises 5.8 points, from an already high 55%. In the high-income households, where more than half of treated mothers worked formally, neither formal nor informal work changes significantly. Adding the mothers' controls makes the estimates somewhat smaller but leaves the pattern in place, and none of the differences between income groups is significant, so the data are consistent with a similar shift in the bottom and middle of the distribution.

Figures 18 and 19 show the same patterns year by year: in Q1 and Q2–Q4, formal work of treated mothers sits about two points above its 2006 level before the policy and slightly below it afterwards, with the decline growing over time in the middle of the distribution, where it reaches about four points by 2024. Informal work, in contrast, moves up after 2007 in Q2–Q4, between three and eight points above 2006, while in Q1 it stays at about the level it already had before the policy, and in Q5 both series move around zero with wide confidence intervals.

Table 10: Heterogeneity by household income, two by two difference in differences

	Household income				
	Q1	Q2–Q4	Q5	p Q2–4	p Q5
Panel A: Baseline					
Works	-0.7†	-1.0†	3.5*	0.81	0.08
	(1.1)	(0.9)	(2.0)		
Searches for work	-2.0	-0.5	-0.7	0.43	0.51
	(1.9)	(0.8)	(1.0)		
Log weekly hours	-0.05	-0.054***	0.01	0.87	0.39
	(0.05)	(0.016)	(0.04)		
Log monthly labor income	-0.07	-0.052***	0.06	0.72	0.04
	(0.05)	(0.019)	(0.04)		
Childcare (inactive, 2006 on)	-2.9	1.8	8.6	0.21	0.28
	(2.8)	(2.8)	(10.5)		
Household chores (inactive, 2006 on)	3.4	1.4	-6.2	0.56	0.64
	(2.8)	(2.0)	(20.0)		
Panel B: All controls					
Works	0.7†	-0.1†	3.7*	0.58	0.19
	(1.0)	(0.8)	(1.9)		
Searches for work	-2.1	-0.3	-0.6	0.34	0.40
	(1.8)	(0.8)	(0.9)		
Log weekly hours	-0.02	-0.050***	0.01	0.64	0.61
	(0.05)	(0.016)	(0.04)		
Log monthly labor income	-0.05	-0.050***	0.06*	0.94	0.08
	(0.05)	(0.017)	(0.04)		
Childcare (inactive, 2006 on)	-2.8	2.0	9.6	0.19	0.23
	(2.9)	(2.6)	(10.3)		
Household chores (inactive, 2006 on)	3.3	1.6	-1.2	0.63	0.79
	(2.9)	(2.1)	(15.8)		
Observations, works (baseline)	66,109	117,689	15,634		
Observations, reasons (inactive women, baseline)	24,777	30,225	1,923		

Notes: Two by two difference in differences by household income, women 22–45, 322 cities, CASEN 1998–2024. Treated: lives with a child 1–4; control: no child 0–4 and a child 6–9. City × year and city × group fixed effects; p of a difference from one regression with every term and fixed effect interacted with the group. Take-up by income, at the child level, is in Table 9. Inactive women: neither employed nor searching for work, the women who are asked the reason; reasons in pp of inactive women, 2006 on (the first wave in which childcare and chores are separable). There is no pre-trend or lead test for outcomes that start in 2006. Each group is estimated separately. p: Q2–Q4 and Q5 against Q1. †: the group's pre-trend test rejects at 5%. Panel B adds the woman's age and years of schooling (linear) and household-role indicators. Full tables in the appendix. Standard errors clustered by city in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 17: Works, event study by household income: Q1 (a), Q2–Q4 (b), Q5 (c)

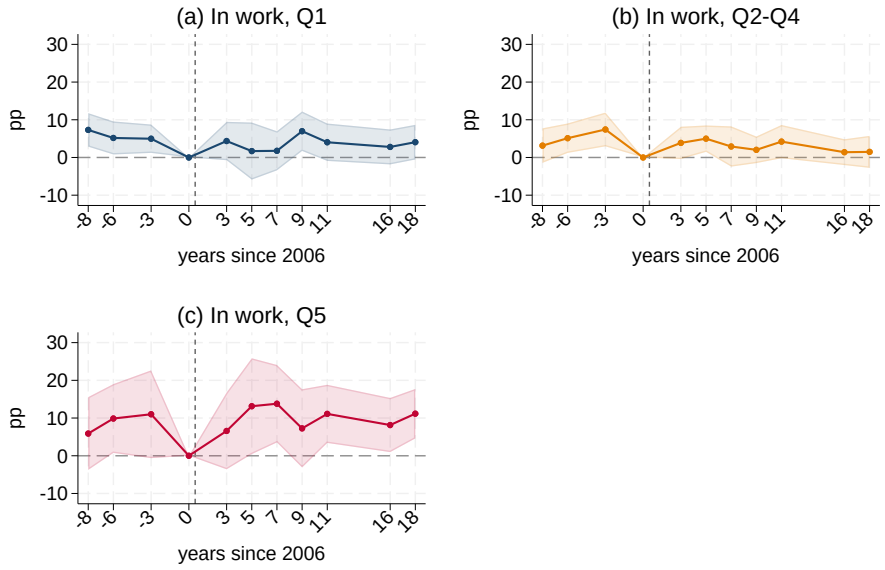


Table 11: Formal and informal work by household income

	Household income				
	Q1	Q2–Q4	Q5	*p* Q2–4	*p* Q5
Panel A: Baseline					
Works, formal (contributes)	-1.9**	-3.6***	2.4	0.16	0.14
	(0.8)	(1.0)	(2.8)		
Works, informal (does not contribute)	1.2	2.4***†	1.2	0.41	0.99
	(1.2)	(0.7)	(2.1)		
Informal share among workers	5.8**	4.0***	0.9	0.55	0.15
	(2.8)	(1.2)	(3.0)		
Panel B: All controls					
Works, formal (contributes)	-1.1	-2.6***	2.7	0.22	0.17
	(0.8)	(0.9)	(2.8)		
Works, informal (does not contribute)	1.8	2.4***†	1.2	0.68	0.77
	(1.2)	(0.6)	(2.0)		
Informal share among workers	3.8	3.3***	0.7	0.87	0.36
	(2.9)	(1.1)	(3.0)		
Observations	65,813	116,738	15,510		

Notes: Formal and informal work by household income (CASEN's national quintile of the household), women 22–45, 322 cities, CASEN 1998–2024. A working woman is formal if she contributes to the pension system and informal if she does not; formal and informal work are in pp of women (they add up to employment), the informal share in pp of working women. Two by two difference in differences, each group estimated separately; *p*: Q2–Q4 and Q5 against Q1, from one regression with every term and fixed effect interacted with the group. †: the group's pre-trend test rejects at 5%. Panel B adds the woman's age and years of schooling (linear) and household-role indicators. Standard errors clustered by city in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 18: Works formal, event study by household income: Q1 (a), Q2–Q4 (b), Q5 (c, own scale)

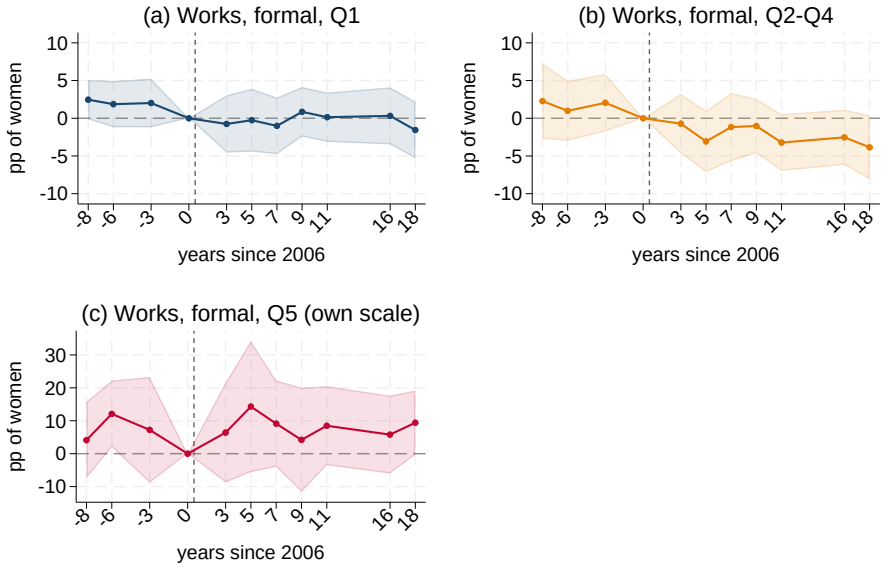
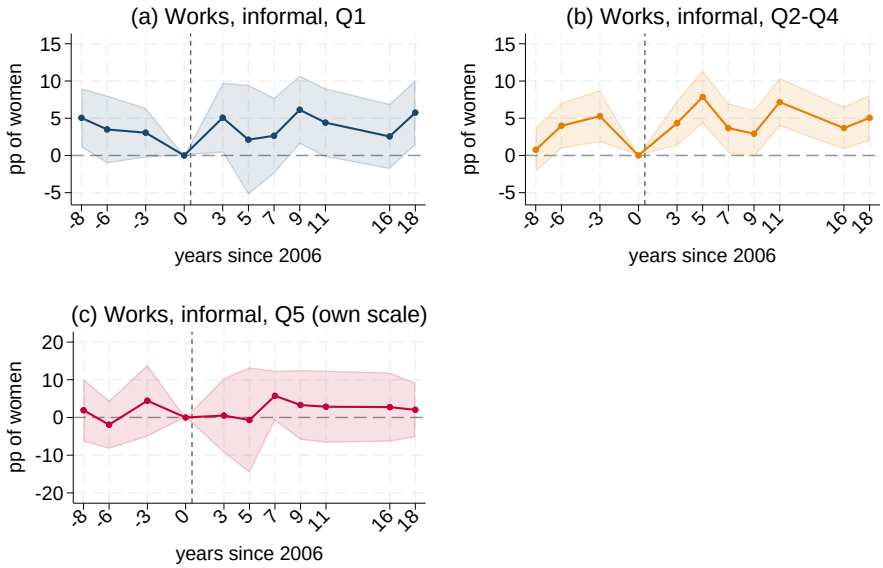


Figure 19: Works informal, event study by household income: Q1 (a), Q2–Q4 (b), Q5 (c, own scale)



Substitution Effects. Substitution across providers follows the same lines (Table 12 and Figures 20 and 21). Measured against its own path before 2006, municipal attendance of treated children ends 7.5 percentage points below that path by 2024 in Q1 (standard error 1.8) and 6.1 points below it in Q2–Q4 (standard error 1.2), large declines compared with municipal attendance of about 4.5 percent in 2006, while in Q5 it does not break (2.1 points below, standard error 2.9; $p < 0.01$ against Q1). Private and voucher care does not fall in any group. JUNJI attendance ends above its old path only in the middle of the distribution, by 4.7 points (standard error 2.1), so in the poorest households the decline of municipal care is not matched by a visible rise of JUNJI above its own trend.

Table 12: Substitution by household income: attendance of treated children relative to the pre-2006 trend, distance in 2024

	Household income				
	Q1	Q2–Q4	Q5	*p* Q2–4	*p* Q5
Panel A: Baseline					
JUNJI	-3.3	4.7**	-3.9	0.10	0.95
	(3.1)	(2.1)	(4.6)		
Public care	-9.1**	0.2	-5.4	0.03	0.68
	(3.9)	(2.8)	(5.6)		
INTEGRA	1.6	1.7	0.6	0.62	0.15
	(2.3)	(1.4)	(2.1)		
Municipal	-7.5***	-6.1***	-2.1	0.05	<0.01
	(1.8)	(1.2)	(2.9)		
Private or voucher	-0.3	-1.1	11.4	0.95	0.16
	(2.6)	(3.0)	(9.0)		
Panel B: All controls					
JUNJI	-4.9	3.6*	-2.6	0.07	0.92
	(3.0)	(2.1)	(4.6)		
Public care	-11.6***	-1.1	-3.4	0.02	0.30
	(3.9)	(2.8)	(5.6)		
INTEGRA	0.9	1.2	0.9	0.72	0.18
	(2.3)	(1.4)	(2.1)		
Municipal	-7.6***	-5.9***	-1.7	0.03	<0.01
	(1.8)	(1.2)	(2.9)		
Private or voucher	-0.5	-2.7	6.9	0.58	0.48
	(2.6)	(2.9)	(8.6)		
Age (linear)	No in Panel A, Yes in Panel B				
Household role (indicators)	No in Panel A, Yes in Panel B				
Years of schooling (linear)	No in Panel A, Yes in Panel B				
Observations	40,860	68,197	9,632		

Notes: Trend break in attendance (pp) of the youngest child aged 1–4 of each treated woman, by group, 322 cities, CASEN 1998–2024. Each cell: distance from the pre-2006 trend in 2024 (jump + 18 × change in slope), estimated separately by group with city fixed effects. Each *p* tests that the jump and the change in slope are the same in the two groups (one regression with every term and fixed effect by group): Q2–Q4 and Q5 against Q1. Survey weights. Observations: children aged 1–4 of treated women in each group, baseline. Standard errors clustered by city in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 20: Municipal attendance of treated children relative to the pre-2006 trend, by household income: Q1 (a), Q2–Q4 (b), Q5 (c)

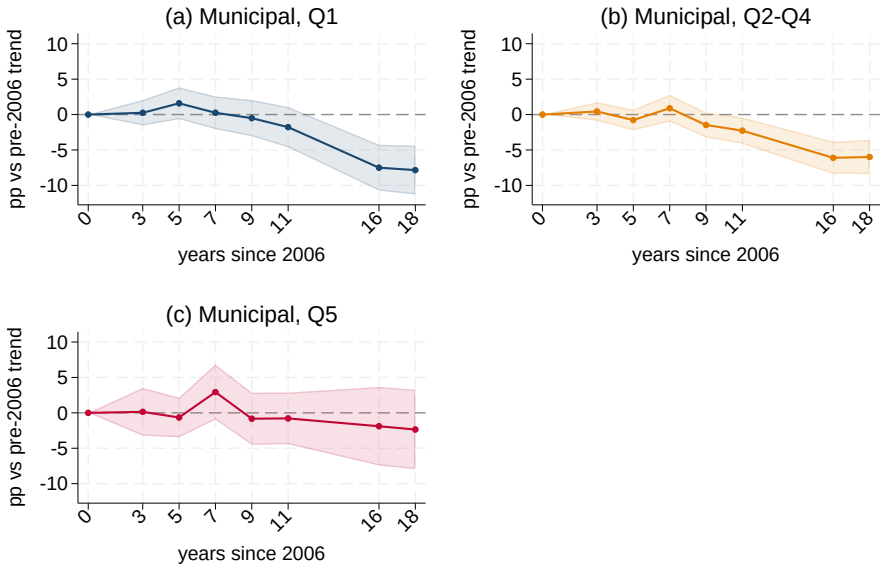
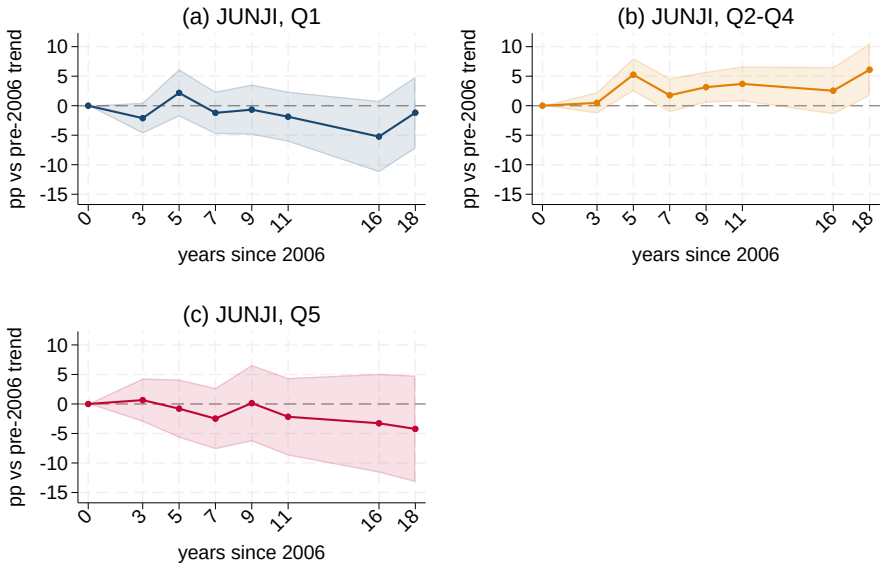


Figure 21: JUNJI attendance of treated children relative to the pre-2006 trend, by household income: Q1 (a), Q2–Q4 (b), Q5 (c)



Time Use. The change in the reasons mothers give for staying out of the labor force is concentrated in the poorest households (Table 13). As a share of all women, mothers of young children in Q1 become 4.1 percentage points less likely to say that childcare is why they do not look for work (standard error 1.9), more than 10% of the 29.5% who said so in 2006. At the same time, the share who cite household chores rises 2.2 points, from 22.9%, and the share who say they are not interested in working does not rise. I do not find significant or large effects of any of these reasons for the other two groups. The event studies show that the shift in Q1 is not driven by a single wave (Figures 22, 23 and 24). The share citing childcare is below its 2006 level in every wave after the policy, by between 2 and 7 points until 2017, while the share citing chores is above it in every wave, by between 1 and 4 points. In Q2–Q4 both series stay close to zero throughout, and in Q5 they are too imprecise to read.

Measuring the reasons among inactive women, the population the question is asked to, removes the spike in the employment of control women in 2006 but halves the precision. With this measure, the Q1 estimates keep their direction, a fall of 2.9 points in childcare and a rise of 3.4 points in chores, from shares of 45% and 34.8%, although neither is significant. No interest in working does not rise in any group under either measure. Taken together, the poorest mothers are the ones who stop citing childcare as the reason they stay at home and move toward household work, and they do not reinterpret their situation as a lack of interest in working.

Figure 22: Childcare as the reason (share of all women), event study by household income: Q1 (a), Q2–Q4 (b), Q5 (c)

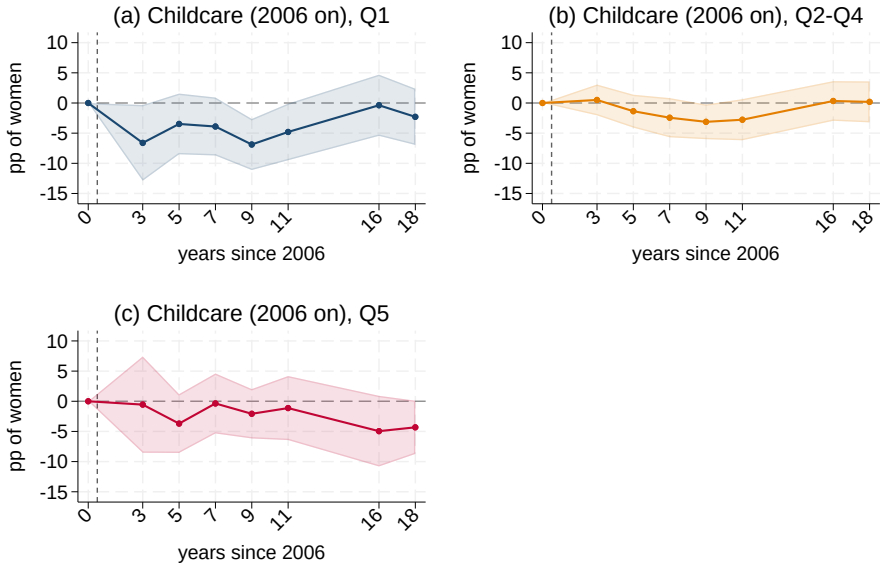


Table 13: Time use: reason for not looking for work by household income

	Household income				
	Q1	Q2–Q4	Q5	p Q2–4	p Q5
Panel A: Share of all women (pp)					
Childcare	-4.1**	-1.2	-2.4	0.20	0.54
	(1.9)	(1.3)	(2.3)		
Household chores	2.2	0.6	-3.5	0.38	0.15
	(1.6)	(0.8)	(3.8)		
No interest in working (all waves)	-0.8***	-0.4*†	-0.3	0.28	0.41
	(0.2)	(0.2)	(0.5)		
Childcare, all controls	-4.4**	-1.4	-2.2	0.15	0.37
	(1.9)	(1.2)	(2.2)		
Household chores, all controls	2.1	0.4	-3.4	0.38	0.17
	(1.7)	(0.8)	(3.7)		
No interest in working (all waves), all controls	-0.8***	-0.4*†	-0.4	0.33	0.53
	(0.2)	(0.2)	(0.6)		
mean 2006, childcare / chores / no interest	29.5 / 22.9 / 2.5	20.0 / 15.2 / 3.0	7.5 / 10.8 / 2.7		
Panel B: Share of inactive women (pp)					
Childcare	-2.9	1.8	8.6	0.21	0.28
	(2.8)	(2.8)	(10.5)		
Household chores	3.4	1.4	-6.2	0.56	0.64
	(2.8)	(2.0)	(20.0)		
No interest in working	-1.3	-2.5**	4.7	0.39	0.39
	(1.0)	(1.1)	(6.9)		
Childcare, all controls	-2.8	2.0	9.6	0.19	0.23
	(2.9)	(2.6)	(10.3)		
Household chores, all controls	3.3	1.6	-1.2	0.63	0.79
	(2.9)	(2.1)	(15.8)		
No interest in working, all controls	-1.3	-2.6**	4.9	0.36	0.38
	(1.0)	(1.1)	(7.1)		
mean 2006, childcare / chores / no interest	45.0 / 34.8 / 3.9	43.4 / 32.8 / 6.5	25.2 / 36.3 / 9.2		
Observations, Panel A (all women)	41,219	82,257	11,387		
Observations, Panel B (inactive women)	24,777	30,225	1,923		

Notes: Reason for not looking for work by household income (CASEN's national quintile of the household), two by two difference in differences, women 22–45, 322 cities, CASEN 2006–2024 (2006 is the first wave in which childcare and chores are separable, so there is no pre-trend test), except no interest as a share of all women, which is comparable across all waves 1998–2024; †: its pre-trend test rejects at 5%. Treated: lives with a child aged 1–4; control: no child aged 0–4 and a child aged 6–9; city × year and city × group fixed effects. Panel A divides by all women, so it also moves with employment, which in 2006 was unusually high among control women; Panel B divides by inactive women (neither working nor searching), the women who are asked the reason. Each group is estimated separately; p: Q2–Q4 and Q5 against Q1, from one regression with every term and fixed effect interacted with the group. All controls adds the woman's age and years of schooling (linear) and household-role indicators. Mean 2006: treated women in 2006 (Panel A) or treated inactive women in 2006 (Panel B). Standard errors clustered by city in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 23: Household chores as the reason (share of all women), event study by household income: Q1 (a), Q2–Q4 (b), Q5 (c)

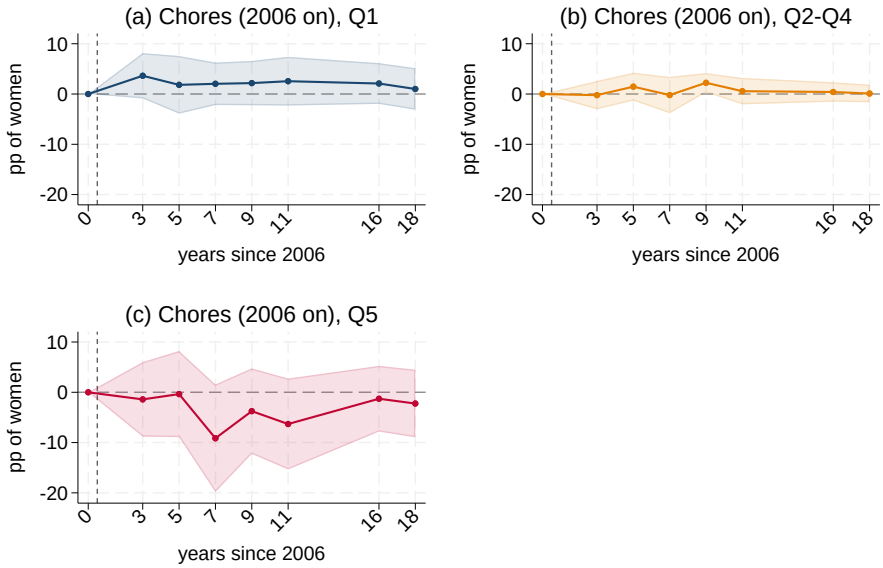
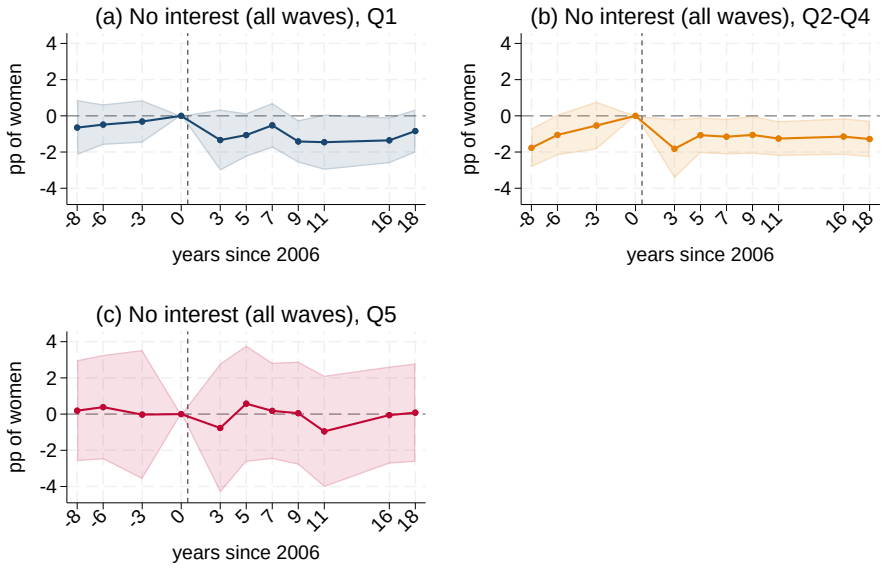


Figure 24: No interest in working as the reason (share of all women, all waves), event study by household income: Q1 (a), Q2–Q4 (b), Q5 (c)



VI.V Robustness: Staggered DID with Imputation

The comparisons between treatment and control groups in the previous subsections identify effects while a mother has a child of daycare age. This definition creates a limitation when the policy has been in place for several years, treatment ends when the child turns 5, even though the policy's effects may persist. Moreover, the control mothers, whose children are aged 6 to 9, were themselves exposed to the treatment earlier (i.e., most of them from 2009 onward, and all of them from 2013 onward). The estimates therefore subtract any effects that persist beyond the daycare years.

To examine these possible persistent effects, I create a new treatment definition where mothers stay treated even when their children are older. A woman becomes exposed once one of her children has been of daycare age in any year after 2007, and remains exposed as the child grows (Section V.IV). The imputation estimator of Borusyak et al. (2024) uses never-exposed and not-yet-exposed women to estimate untreated outcome paths, predicts counterfactual outcomes for exposed women, and averages the differences between observed and predicted outcomes. This approach distinguishes effects for mothers currently exposed, whose youngest child is aged 1 to 4, from effects for previously exposed mothers, whose youngest child is aged 6 to 9.

Childcare take-up. At the child level, the imputation compares exposed children of 1 to 4 years old with children who were never exposed. Table 14 shows that JUNJI attendance increases by 6.4 to 8.1 percentage points, depending on whether newborns are included among the unexposed. These estimates fall within the range reported in Section VI.I. However, the lead test rejects because JUNJI attendance was already increasing before 2006. I therefore continue to rely on enrollment in the new facilities to quantify take-up, since that measure does not depend on the preexisting trend.

Figure 25 shows the effect by survey year. JUNJI attendance of exposed children is 3.4 percentage points above its predicted level in 2009, between 6.5 and 9.9 percentage points from 2011 to 2022, and 13.7 points in 2024, so we can think that it grows with the new facilities. In 2006, however, the children who would later be exposed already attended JUNJI 1.5 points more than predicted, which is the reason why the lead test failed.

Labor market outcomes and type of work. Table 15 reports the staggered estimates for mothers' outcomes, with labor market outcomes in Panel A and reasons for being inactive in Panel B. The employment estimates do not show a credible increase. Employment is 2.8 percentage points above its predicted level among currently exposed mothers and 2.5 points above it among previously exposed mothers, but the lead test rejects ($p < 0.01$). Moreover, Figure 26 shows a positive gap of approximately the same magnitude at every age of the youngest child, from 1 to 10.

Changes in the type of work follow the same timing (Figure 27). Currently exposed mothers

are 2.9 percentage points more likely than predicted to work informally (standard error 0.7), with estimates ranging from 2.8 to 3.0 points across specifications. The age-specific gap ranges from 2 to 3.5 points while the youngest child is of 0 to 4 years old and it is very small from age 6 to 10. Formal employment does not decline among currently exposed mothers (-0.2 points) and it is slightly above its predicted level once the child is older. I interpret that result as suggestive since the pre-policy test for informal employment yields $p = 0.04$.

Time use. These results are consistent with previous findings, even have larger effects (Panel B of Table 15). Relative to never-exposed mothers, currently exposed inactive mothers are 2.9 percentage points less likely than predicted mothers to use childcare (standard error 1.6) as their reason for being out of the labor force. At the same time, they are 4.5 points more likely to indicate household chores (standard error 1.6) as their reason. These changes correspond to approximately 7 and 13 percent of their 2006 levels, respectively. Controlling for the mother's age, schooling, and household role leaves the decline in childcare between 2.8 and 3.0 points, significant at the 10 percent level in every specification, and the increase in chores between 4.0 and 5.0 points.

The age profiles reveal different persistence across the two outcomes. In panels c and d of Figure 26, the decline in childcare is present at every age of the youngest child, ranging from 2 to 7 points. The increase in chores is largest at age 1, at approximately 8 points, and diminishes as the child grows. The decline in childcare persists beyond school entry: mothers whose youngest child is aged 6 to 9 remain 3.4 percentage points less likely to cite childcare (standard error 1.5). There is no corresponding increase in a reported lack of interest in working, which does not change at any age.

These findings suggest that the policy changed how mothers who stay out of the labor force described the constraints they faced. Childcare became less usual, while household work became more common. This pattern is consistent with the model's prediction that a non-working mother may enroll her child in daycare without entering employment, reallocating time previously spent with the child toward household work (Section II) or leisure.

Analysis by Income Table 16 presents the outcomes by three income groups. Employment does not respond in Q1, the only group whose lead test passes (1.9 points, standard error 1.7), and the fall in hours comes again from the middle of the distribution. The move into informal work while the child is small appears in every income group, between 2.0 and 2.9 points, although it is precise only in Q2–Q4. Among inactive mothers, chores rise by about 5 points in Q1 and in Q2–Q4, significant in both, while childcare falls by about 3 points in each group, with less precision (Figures 28 and 29); Q5 has too few inactive mothers to say anything.

Two features of the estimator call for caution. Mothers of infants count as not yet exposed, so if they change what they do while they wait for a place, they are not a clean picture of life without the policy. And because effects are indexed by the age of the youngest child while a mother can be

exposed through an older sibling, they describe mothers by the age of their youngest child rather than strictly by the time since they were first exposed. Finally, the reasons start in 2006, so there is no test for differences before the policy for them. With these caveats in mind, the staggered version completes the story of this section. The expansion moved children between public providers, left mothers' employment where it was, and shortened the hours of working mothers while their children were small. For the mothers who stayed at home, it changed the reason they give for being there, from the care of the child to the work of the house, and that change lasted.

Table 14: Staggered version: take-up at the child level, imputation estimator

	JUNJI, children 1–4	Lead test p
Unexposed: children born 2002 or before	6.4***	<0.01
	(0.5)	
Unexposed: + infants	8.1***	<0.01
	(0.4)	

Notes: Imputation at the child level, children 0–12 in 322 cities, CASEN 1998–2024. Outcome: JUNJI attendance (pp) for children 0–4 and attendance at any establishment for older children. Fixed effects: city × year and child age, estimated on unexposed children. The effect averages exposed children aged 1–4. Lead test: children observed in 2003 and 2006 who will be 1–4 after 2007. Standard errors clustered by city in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 25: Staggered version: JUNJI attendance of exposed children aged 1–4 by survey year, imputation estimator; 2006 is the lead coefficient (children who will be exposed after 2007)

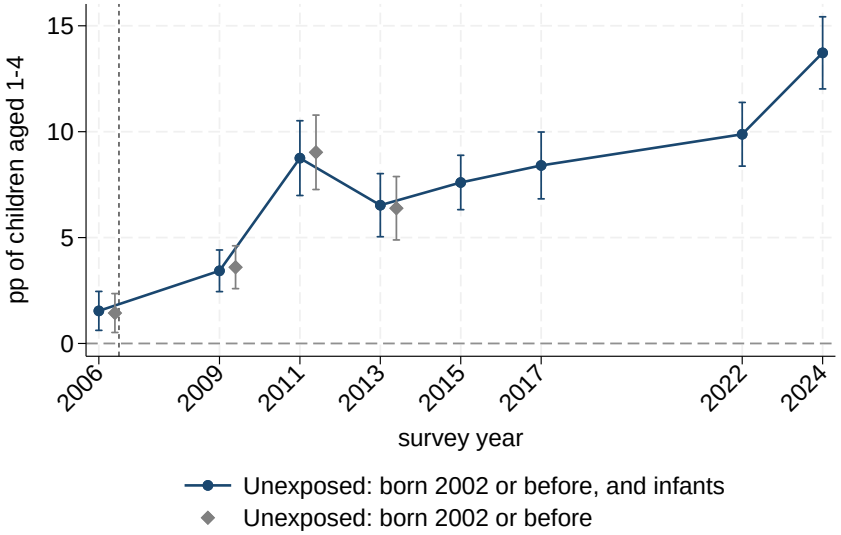


Table 15: Staggered version, imputation estimator: work and reasons for not looking for work

	Youngest child	(1)	(2)	(3)	(4)	(5)
Panel A: Work (women 22–45, 1998–2024)						
Works	1–4 (now)	2.84**	2.76**	2.25**	3.09***	2.25**
		(1.122)	(1.112)	(1.109)	(1.096)	(1.047)
lead p = <0.01	6–9 (before)	2.46**	2.41**	1.13	2.61**	0.98
		(1.172)	(1.163)	(1.197)	(1.191)	(1.193)
Searches for work	1–4 (now)	-1.10	-1.02	-1.19	-1.16	-1.13
		(0.789)	(0.796)	(0.804)	(0.783)	(0.795)
lead p = 0.28	6–9 (before)	0.83	0.82	0.58	0.81	0.69
		(0.705)	(0.691)	(0.702)	(0.708)	(0.695)
Log weekly hours	1–4 (now)	-0.04**	-0.04**	-0.04**	-0.04**	-0.04**
		(0.015)	(0.015)	(0.015)	(0.015)	(0.015)
lead p = 0.10	6–9 (before)	-0.01	-0.01	-0.01	-0.01	-0.01
		(0.019)	(0.019)	(0.018)	(0.019)	(0.019)
Log monthly labor income	1–4 (now)	-0.05**	-0.05**	-0.05**	-0.03*	-0.03*
		(0.022)	(0.021)	(0.021)	(0.019)	(0.019)
lead p = 0.77	6–9 (before)	-0.01	-0.01	-0.01	0.00	0.00
		(0.024)	(0.023)	(0.023)	(0.022)	(0.021)
Works, formal	1–4 (now)	-0.16	-0.20	-0.64	0.07	-0.59
		(1.179)	(1.174)	(1.138)	(1.105)	(1.038)
lead p = 0.15	6–9 (before)	1.82*	1.80*	0.73	1.96*	0.67
		(1.045)	(1.041)	(1.031)	(1.053)	(1.025)
Works, informal	1–4 (now)	2.95***	2.90***	2.83***	2.98***	2.81***
		(0.675)	(0.673)	(0.681)	(0.652)	(0.656)
lead p = 0.04	6–9 (before)	0.62	0.59	0.34	0.65	0.28
		(0.617)	(0.618)	(0.636)	(0.606)	(0.621)
Informal share of workers	1–4 (now)	1.42	1.39	1.57	1.18	1.33
		(1.260)	(1.266)	(1.253)	(1.202)	(1.186)
lead p = 0.48	6–9 (before)	-2.05**	-2.04**	-1.62	-1.94**	-1.52
		(0.971)	(0.980)	(1.003)	(0.946)	(0.964)
Panel B: Reason for not looking for work (pp of inactive women, 2006–2024)						
Childcare	1–4 (now)	-2.86*	-2.85*	-2.89*	-2.96*	-2.90*
		(1.636)	(1.640)	(1.603)	(1.653)	(1.610)
lead p = –	6–9 (before)	-3.41**	-3.37**	-3.37**	-3.54**	-3.15**
		(1.458)	(1.465)	(1.450)	(1.502)	(1.504)
Household chores	1–4 (now)	4.47***	4.22**	5.02***	4.05**	4.58***
		(1.625)	(1.671)	(1.745)	(1.653)	(1.706)
lead p = –	6–9 (before)	2.64*	1.43	3.81**	2.14	2.57*
		(1.584)	(1.522)	(1.584)	(1.525)	(1.502)
No interest in working	1–4 (now)	-0.32	-0.31	-0.28	-0.30	-0.26
		(1.178)	(1.177)	(1.184)	(1.197)	(1.200)
lead p = –	6–9 (before)	0.33	0.35	0.38	0.38	0.43
		(0.806)	(0.807)	(0.812)	(0.816)	(0.815)
Age (linear)		No	Yes	No	No	Yes
Household role (indicators)		No	No	Yes	No	Yes
Years of schooling (linear)		No	No	No	Yes	Yes
Exposed women, youngest 1–4 (works)		60,765	60,765	60,765	60,609	60,609
Exposed inactive women, youngest 1–4 (reasons)		25,157	25,157	25,157	25,105	25,105

Notes: Imputation estimator (Borusyak et al., 2024), women 22–45 with a child 0–17, 322 cities, CASEN. Average gap between exposed women and their predicted outcome without the policy, by the age of the youngest child (1–4 now, 6–9 before). Formal: contributes to a pension system; Panel B: pp of inactive women, 2006 on (no lead test). FE: city × year, youngest-child age and number of children; (2)–(5) add age and schooling (linear) and role indicators. Lead p: column (1). Observations: exposed women imputed. Standard errors clustered by city in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 26: Imputation: effect by the age of the youngest child; works (a) and log weekly hours (b) of all women, childcare (c) and household chores (d) as the reason, share of inactive women

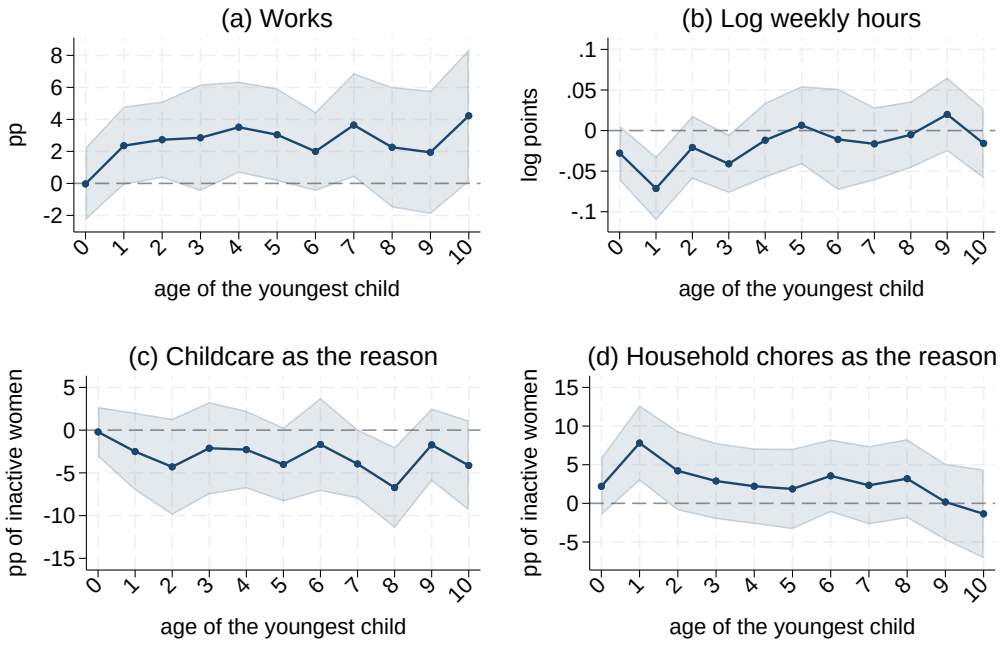


Figure 27: Formal and informal work, imputation by the age of the youngest child: works formal (a), works informal (b)

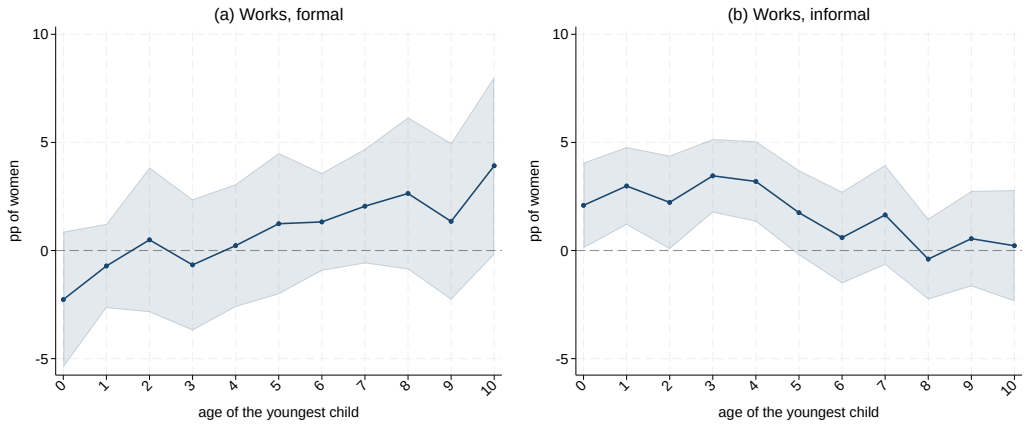


Table 16: Staggered version by household income

		Household income		
	Youngest child	Q1	Q2–Q4	Q5
Works	1–4 (now)	1.9	3.9***†	0.9†
		(1.7)	(1.4)	(2.4)
Log weekly hours	1–4 (now)	0.01	-0.037**	-0.02
		(0.06)	(0.014)	(0.03)
	6–9 (before)	-0.02	0.003	-0.01
		(0.06)	(0.020)	(0.03)
Works, informal	1–4 (now)	2.0	2.5***†	2.9*
		(1.3)	(0.7)	(1.7)
	6–9 (before)	0.5	0.4†	1.7
		(1.3)	(0.7)	(2.2)
Reason for not looking for work (pp of inactive women)				
Childcare	1–4 (now)	-3.4	-2.5	-3.1
		(2.4)	(2.5)	(6.0)
	6–9 (before)	-1.5	-3.3	-6.0
		(2.9)	(2.1)	(4.8)
Household chores	1–4 (now)	5.1**	5.6**	-2.3
		(2.2)	(2.6)	(8.2)
	6–9 (before)	0.6	3.4	8.8
		(3.0)	(2.6)	(8.4)
Exposed women, youngest 1–4 (works)		15,969	36,229	4,968
Exposed inactive women, youngest 1–4 (reasons)		9,067	12,962	747

Notes: Staggered version (imputation) by household income (CASEN's national quintile of the household), baseline fixed effects as in Table 15; each group imputed separately. Works in pp; hours in logs for women who work; informal work in pp of women; reasons in pp of inactive women, 2006 on. †: the group's lead test rejects at 5%; there is no lead test for the reasons. Standard errors clustered by city in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 28: Childcare as the reason (share of inactive women), imputation by the age of the youngest child and household income: Q1 (a), Q2–Q4 (b)

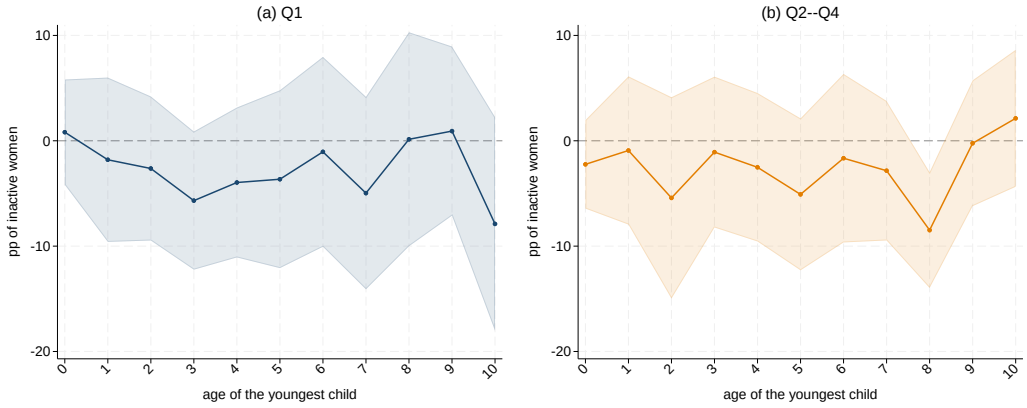
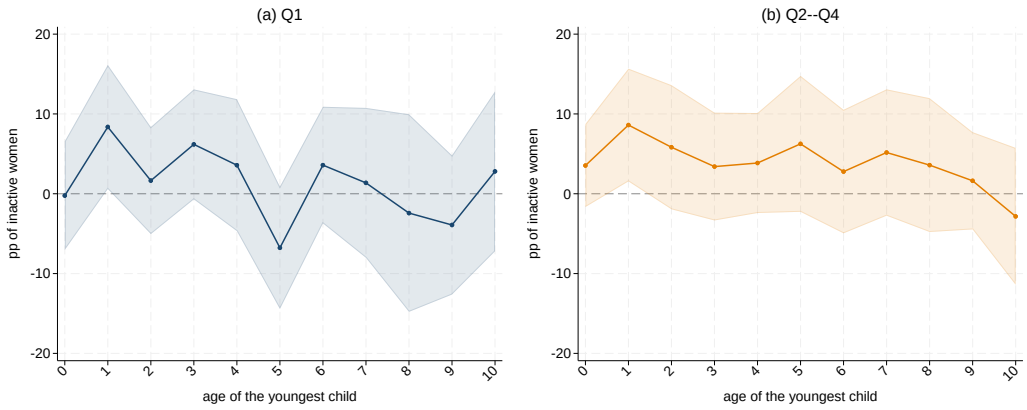


Figure 29: Household chores as the reason (share of inactive women), imputation by the age of the youngest child and household income: Q1 (a), Q2–Q4 (b)



VII Conclusion

Public childcare policies aim to reduce mothers' childcare responsibilities and facilitate their participation in the labor market. However, the evidence on whether these policies increase maternal employment is mixed. In this paper, I evaluate Chile's national expansion of public childcare through JUNJI, which started in 2007 as part of Chile Crece Contigo program. I use more than 25 years of survey data and administrative records to estimate the effects of the expansion on childcare attendance, mothers' labor market outcomes, and their decisions on how to allocate their time. Because the expansion occurred across the country, I use the age of the children living with each mother to define eligibility and estimate a difference-in-differences approach.

I find that the expansion significantly increased JUNJI enrollment, but did not lead to an increase in maternal employment. By 2024, facilities opened after 2006 enrolled 9.8 children per 100 children under five. However, this increase was accompanied by a decline in municipal daycare enrollment, suggesting that part of the expansion replaced existing public childcare rather than increasing total coverage. In terms of labor market outcomes, I find no significant increase in mothers' employment across income groups. Instead, formal employment decreases while informal employment increases, and working mothers reduce their weekly hours by approximately 5% while their children are of daycare age.

I examine two possible explanations for these results. First, mothers who were already using municipal daycare could move their children to JUNJI without gaining additional time to work. By 2024, municipal enrollment was 6.2 children per 100 below its pre-policy trend, offsetting much of the increase in JUNJI enrollment. Second, childcare is not the only constraint mothers face when deciding whether to work. Among inactive mothers exposed to the policy, the probability of reporting childcare as their reason for not looking for a job decreases by approximately 3 percentage points, while reporting household chores increases by 4.5 percentage points. This suggests

that even when childcare becomes available, other household responsibilities may continue to limit mothers' labor force participation.

To understand these results, I develop a simple model in which mothers choose their childcare provider, whether to work, and how to allocate their time between childcare, household work, and leisure. The model shows that increasing access to a preferred provider can raise enrollment without increasing employment. Mothers who switch between providers do not gain additional time, while mothers who previously cared for their children at home may use the new daycare and still choose not to work. In this case, the time previously allocated to childcare can be reallocated to household work and leisure. The model therefore provides an explanation for why childcare attendance increases while maternal employment remains unchanged.

An additional consideration is how the effects of long-running childcare policies are estimated. The standard difference-in-differences approach compares mothers of daycare-age children with mothers of older children. However, as the policy continues over time, mothers in the control group may have been previously exposed to the expansion. To address this issue, I complement the standard estimation with the imputation estimator proposed by Borusyak et al. (2024). This distinction is particularly important for the reasons mothers give for not looking for work, where the staggered approach identifies persistent effects that the standard estimation does not capture.

These findings have important implications for childcare policies. First, an increase in public daycare enrollment does not necessarily represent an equivalent increase in childcare coverage, since families may substitute between existing providers. Targeting new facilities toward families without access to childcare may therefore be more effective in increasing coverage. Second, expanding daycare availability alone may not be sufficient to increase maternal employment, particularly among low-income mothers. The results suggest that other household responsibilities continue to influence mothers' labor market decisions even when childcare becomes available. Therefore, policies that aim to increase female labor force participation need to consider these barriers together with the provision of childcare.

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A Triple Difference: Fathers as an Additional Comparison

The main comparison could pick up anything that happened after 2006 to parents of young children and not to parents of school-age children, for example a change in local labor demand for workers with small children. To check this, I use fathers. I estimate a triple difference that compares the gap between mothers of children aged 1 to 4 and mothers of children aged 6 to 9 with the same gap for fathers, before and after the policy. The regression includes city $\times$ year fixed effects separately for each sex, so it absorbs the long-run convergence of mothers' and fathers' employment in Chile, which is large in these years. Using fathers of children aged 1 to 4 directly as the control group does not work for this reason: the event study for employment is a straight line from 1998 to 2024, which reflects that convergence and not the policy.

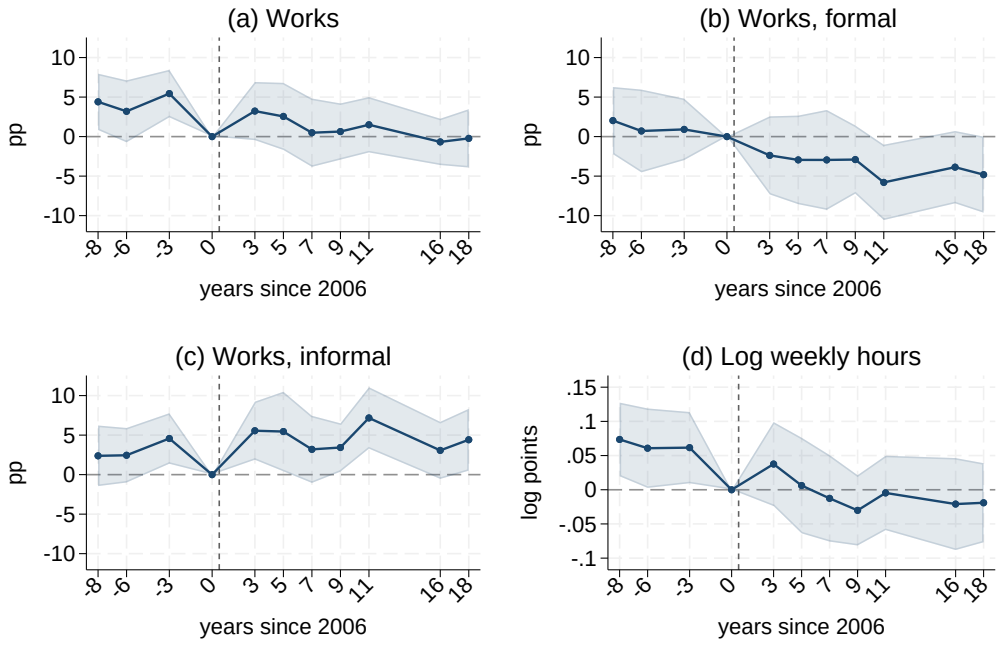
Table 17 and Figure 30 show the results. Employment does not increase. The estimate is -2.2 percentage points (standard error 0.9) and -1.1, not significant, with all the controls, and as in the main results the pre-trend test rejects because of the unusual 2006 wave. Hours and income fall about 5 percent, with a pre-trend test that passes for income. The fall in formal work is clearer than in the main design: formal work decreases 4.6 points (standard error 1.2), the years before 2006 are flat ($p = 0.79$), and the coefficients are between -2.4 and -5.8 points from 2009 on. Informal work increases 2.2 points (standard error 0.9). Childcare as the reason for not looking for work falls 2.0 points (standard error 1.0), and the estimate stays the same with all the controls. Fathers almost never give this reason, so this last result works mainly as a check that it does not come from a shock common to all parents. Overall, subtracting fathers does not change the main results, and it makes the shift from formal to informal jobs more credible.

Table 17: Triple difference: mothers against fathers, children 1–4 against children 6–9

	Baseline	All controls	Pre-trend p
Works	-2.2**	-1.1	<0.01
	(0.9)	(0.9)	
Searches for work	-0.1	-0.2	0.41
	(0.8)	(0.8)	
Log weekly hours	-0.054***	-0.052***	0.06
	(0.014)	(0.014)	
Log monthly labor income	-0.054***	-0.046**	0.41
	(0.018)	(0.018)	
Works, formal	-4.6***	-3.5***	0.79
	(1.2)	(1.2)	
Works, informal	2.2**	2.4***	0.05
	(0.9)	(0.9)	
Childcare as the reason (2006 on)	-2.0**	-2.1**	–
	(1.0)	(1.0)	
Household chores as the reason (2006 on)	0.4	0.3	–
	(0.8)	(0.8)	
No interest in working	-0.53***	-0.57***	0.14
	(0.17)	(0.17)	
Observations (first row)	352,938	351,491	

Notes: Triple difference, CASEN 1998–2024, 322 cities: women and men 22–45 living with a child 0–17. Each cell is the coefficient on treated × woman × post-2006, where treated means living with a child aged 1–4 and the comparison within each sex is having no child aged 0–4 and a child aged 6–9. Fixed effects: city × year × sex, city × group × sex, and year × group. Works, search, formal and informal work and the reasons in pp; hours and income in logs for those who work. Pre-trend p: joint test of the 1998, 2000 and 2003 event-study coefficients (– when the outcome starts in 2006). All controls adds the woman's age and years of schooling (linear) and household-role indicators. Standard errors clustered by city in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 30: Triple difference, event study relative to 2006: works (a), works formal (b), works informal (c), log weekly hours (d)



B Mothers of Children Aged 6 to 7 as the Control Group

The control group of the main analysis includes mothers of children up to 9 years old. Here I restrict it to mothers whose children are 6 or 7, the closest ages to the treated children, so the two groups are as similar as possible in their stage of life, and the control children are below the ages of the full-time school policy. The control group is smaller, so the estimates are less precise (Table 18 and Figure 31).

The results are very close to the main ones. Employment does not change (-0.4 points, standard error 0.8, and 0.3 with all the controls), and the event study shows the same pattern around 2006 as the main sample. Formal work falls 2.6 points (standard error 0.8) and informal work rises 2.1 points (standard error 0.7), so the change in the type of work does not depend on the age of the control children. Hours and income fall about 3 percent, significant only at the 10 percent level. For time use, childcare as the reason falls 2.2 points as a share of all women, significant at the 10 percent level, and among inactive women household chores rise 2.8 points, not significant, while no interest in working falls.

Figure 31: Mothers of children 6–7 as the control group, event study relative to 2006: works (a), works formal (b), works informal (c), childcare as the reason (d)

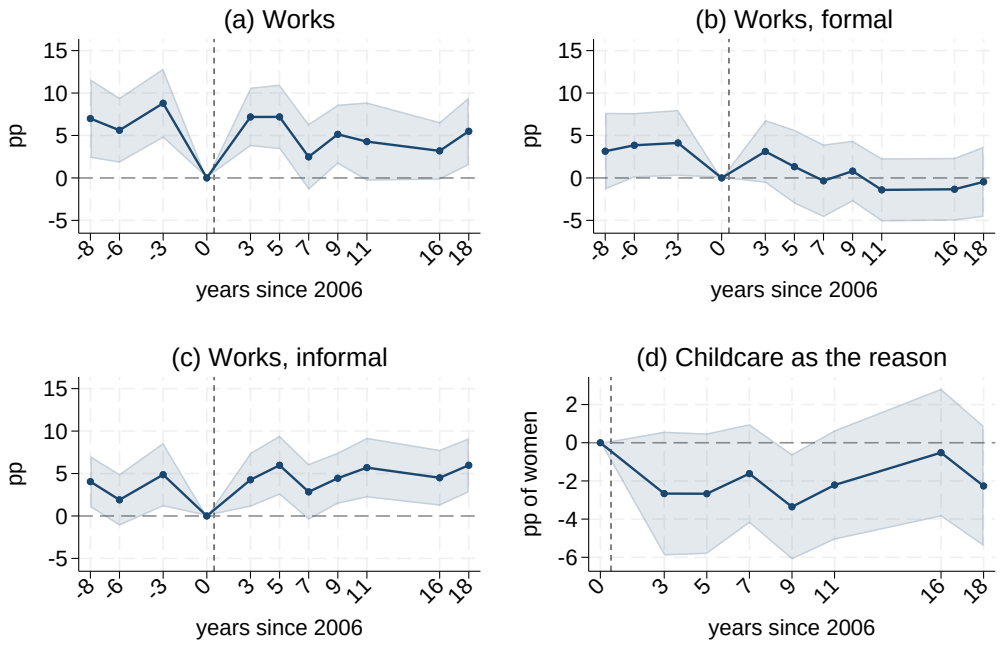


Table 18: Mothers of children 6–7 as the control group, two by two difference in differences

	Baseline	All controls	Pre-trend p
Works	-0.4	0.3	<0.01
	(0.8)	(0.8)	
Searches for work	-1.0	-1.0	0.08
	(0.8)	(0.8)	
Log weekly hours	-0.030*	-0.024	0.10
	(0.018)	(0.018)	
Log monthly labor income	-0.030*	-0.024	0.10
	(0.016)	(0.016)	
Works, formal	-2.6***	-2.0**	0.14
	(0.8)	(0.9)	
Works, informal	2.1***	2.2***	0.03
	(0.7)	(0.7)	
Childcare as the reason (2006 on)	-2.2*	-2.3*	–
	(1.3)	(1.3)	
Household chores as the reason (2006 on)	0.2	0.2	–
	(1.0)	(0.9)	
Childcare as the reason, inactive (2006 on)	-0.1	-0.3	–
	(2.8)	(2.8)	
Household chores as the reason, inactive (2006 on)	2.8	2.6	–
	(1.9)	(1.9)	
No interest, inactive (2006 on)	-1.9**	-1.9**	–
	(0.9)	(0.9)	
Observations (first row)	163,181	162,551	

Notes: Two by two difference in differences, women 22–45, 322 cities, CASEN 1998–2024. Treated: lives with a child aged 1–4; control: no child aged 0–4 and a child aged 6–7. City × year and city × group fixed effects. Reasons in pp of all women or, where indicated, of inactive women (neither working nor searching). Pre-trend p: joint test of the 1998, 2000 and 2003 event-study coefficients (– when the outcome starts in 2006). All controls adds the woman's age and years of schooling (linear) and household-role indicators. Standard errors clustered by city in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C Dropping the 2006 Survey Wave

In 2006, the last wave before the policy, employment of control women is unusually high (Figure 11). Since every event-study coefficient is measured against 2006, this year can move the results. Here I drop the 2006 wave and compare every year after the policy with the average of 1998–2003. The reasons for not looking for work cannot be estimated this way, because they only exist from 2006 on.

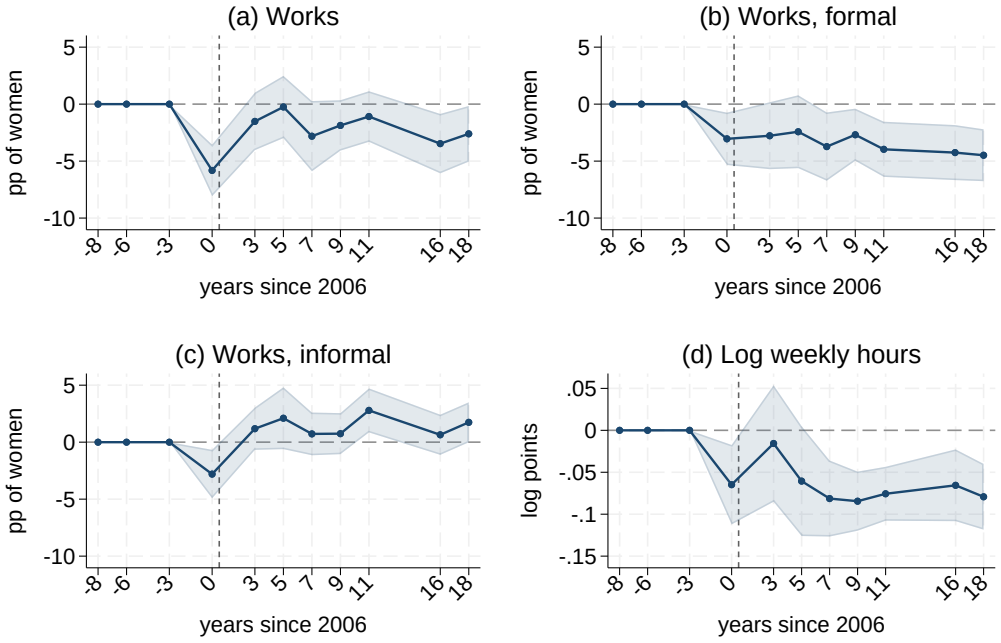
Table 19 and Figure 32 show that, against 1998–2003, the years before the policy are flat for every outcome. Employment falls 2.0 points (standard error 0.8), and 1.1 points, not significant, with all the controls, so in no case the policy increased employment. Formal work falls 3.5 points with a flat pre-period, informal work rises 1.4 points, and hours fall 6.7 percent. The last columns of the table show that 2006 was an unusual year for all these outcomes. Employment and informal work dip in 2006 and go back up after the policy, while formal work and hours are already at their lower level in 2006, so this comparison cannot date when these two changes started. For this reason, I rely on the triple difference to read the timing of the changes in hours and in the type of work.

Table 19: Labor market against 1998–2003: two by two without the 2006 wave, and the gap in 2006

	Two by two, reference 1998–2003			Gap in 2006	
	Baseline	All controls	Pre-trend p	Baseline	All controls
Works	-2.0**	-1.1	0.33	-5.8***	-5.0***
	(0.8)	(0.8)		(1.2)	(1.0)
Works, formal	-3.5***	-2.7***	0.70	-3.0***	-2.1*
	(0.9)	(0.8)		(1.2)	(1.1)
Works, informal	1.4**	1.6***	0.13	-2.8***	-2.9**
	(0.6)	(0.6)		(1.1)	(1.2)
Informal share of workers	3.5***	2.7***	0.21	-1.2	-2.2
	(1.1)	(1.0)		(2.0)	(2.2)
Searches for work	-0.5	-0.3	0.32	1.5	1.6
	(0.9)	(0.9)		(1.3)	(1.2)
Log weekly hours	-0.067***	-0.062***	0.71	-0.06***	-0.06***
	(0.015)	(0.015)		(0.02)	(0.02)
Log monthly labor income	-0.040**	-0.039**	0.62	-0.02	-0.02
	(0.020)	(0.018)		(0.04)	(0.03)
Observations (works)	179,258	178,528			

Notes: Women 22–45, 322 cities, CASEN 1998–2024. Two by two difference in differences without the 2006 wave, so the pre-policy period is 1998–2003: treated (lives with a child aged 1–4) against control (no child aged 0–4 and a child aged 6–9), city × year and city × group fixed effects. Pre-trend p: joint test of 1998 and 2000 relative to 2003, baseline. Gap in 2006: event-study coefficient of 2006 relative to the 1998–2003 average. A working woman is formal if she contributes to the pension system and informal if she does not; formal and informal work are in pp of women (they add up to employment), the informal share in pp of working women. Search is defined for women in the labor force; hours and income in logs for women who work. All controls adds the woman's age and years of schooling (linear) and household-role indicators. Standard errors clustered by city in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 32: Labor market, event study relative to 1998–2003: works (a), works formal (b), works informal (c), log weekly hours (d)



D Local Supply of JUNJI Daycares

The main specification compares treated and control women inside the same city and year. A concern is that cities that received more JUNJI facilities were also changing in other ways. Here I replace the city $\times$ year fixed effects with fixed effects for the decile of JUNJI facilities per 1,000 children aged 0 to 4 of each city in each year. Treated and control women are then compared inside the same level of local JUNJI supply, and not only inside the same city.

The results do not change (Table 20 and Figure 33). The JUNJI attendance of the youngest child rises 9.3 points in both specifications. Employment changes by -0.5 points and search by -1.0, none of them significant. Hours fall 4.9 percent and income 3.5 percent, as in the main results. Childcare as the reason for not looking for work falls 1.8 points (standard error 1.0), against 1.9 in the main specification. Therefore, the results do not come from differences in the supply of JUNJI between the cities where treated and control women live.

Table 20: Local supply of JUNJI: supply-decile $\times$ year fixed effects

	Main specification	Supply-decile $\times$ year FE	Pre-trend p
Youngest child 1–4 attends JUNJI	9.3***	9.3***	<0.01
	(0.4)	(0.4)	
Works	-0.5	-0.5	<0.01
	(0.7)	(0.7)	
Searches for work	-0.8	-1.0	0.37
	(0.7)	(0.7)	
Log weekly hours	-0.049***	-0.049***	0.12
	(0.014)	(0.013)	
Log monthly labor income	-0.037**	-0.035**	0.88
	(0.015)	(0.015)	
Childcare as the reason (2006 on)	-1.9*	-1.8*	–
	(1.0)	(1.0)	
Household chores as the reason (2006 on)	0.4	0.1	–
	(0.8)	(0.8)	
No interest in working	-0.51***	-0.47***	0.06
	(0.16)	(0.16)	
Observations (works)	200,759	200,761	

Notes: Two by two difference in differences, women 22–45, 322 cities, CASEN 1998–2024; treated: lives with a child aged 1–4; control: no child aged 0–4 and a child aged 6–9. Main specification: city $\times$ year and city $\times$ group fixed effects. Supply column: the city $\times$ year fixed effects are replaced by fixed effects for the decile of JUNJI facilities per 1,000 children aged 0–4 of the city in that year, interacted with the year, so treated and control women are compared within the same level of local JUNJI supply. The first row is the JUNJI attendance of the woman's youngest child of the group's age. Pre-trend p of the supply specification: joint test of the 1998, 2000 and 2003 event-study coefficients (– when the outcome starts in 2006). Standard errors clustered by city in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 33: Supply-decile × year fixed effects, event study relative to 2006: works (a), log weekly hours (b), childcare as the reason (c), household chores as the reason (d)



E Fertility and the Supply of JUNJI

The design compares women by the age of their children, so it could be biased if the policy changed who has children. If daycare became cheaper, some women could decide to have a child, and the treated women after 2006 would be different from the treated women before it. I test this in three ways (Table 21). Since every woman of a city faces the same daycare supply, these estimates compare cities where the expansion was large relative to their initial supply (under-served cities) with the rest.

None of the three sources shows a response. In the household survey, the number of children aged 0 to 4 living with women aged 25 to 45 changes by 0.010 (standard error 0.013), and the probability of having a child under one by 0.169 points (standard error 0.462), over a base of 6.0%. In the vital statistics, where I observe every registered live birth by the city of residence of the mother, the effect on log births is −0.000 (standard error 0.015). The INE population projections give the same null, although their pre-trend tests reject because they are re-anchored at each census. Figure 34 shows that births in the cities of my sample fall from 2014 on, while JUNJI supply keeps growing.

Figure 34: Live births and JUNJI facilities per 100 children aged 0–4, cities of the sample, 2000–2020

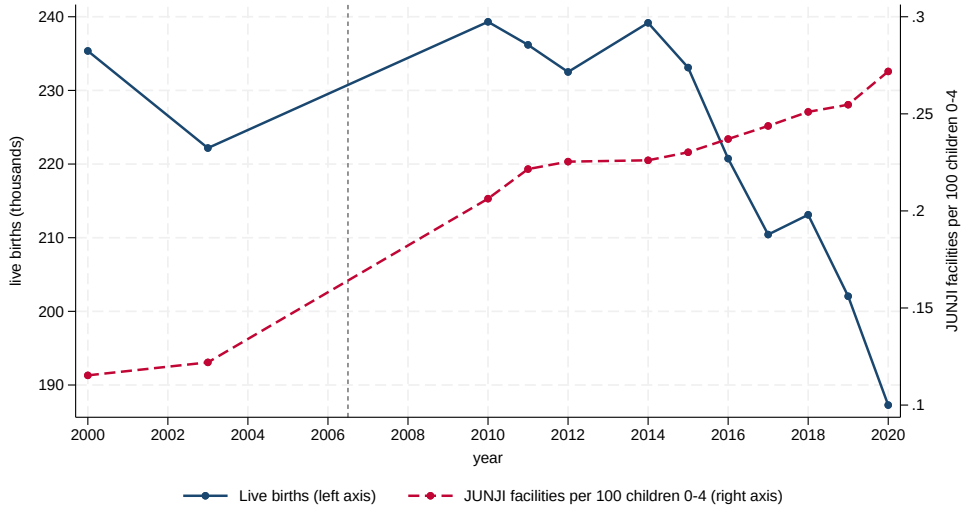


Table 21: Fertility and the expansion of JUNJI

	Treatment effect	Pre-trend p	Control mean	Obs.
Panel A. Household survey (CASEN), women aged 25 to 45				
Children aged 0–4 in the household	0.010	0.275	0.35	334,740
	(0.013)			
Children aged 1–4 in the household	0.008	0.358	0.29	334,740
	(0.011)			
Has a child under one (pp)	0.169	0.881	5.96	334,740
	(0.462)			
All children in the household[a]	0.147***	0.627	2.23	334,740
	(0.051)			
Panel B. Registered live births, by city of residence of the mother				
Log live births	-0.000	0.889	6.69	3,791
	(0.015)			
Live births (count)	108.2	0.563	1,495.35	3,792
	(89.5)			
Panel C. INE population projections, population aged under one				
Log births	0.005	0.000	7.15	7,337
	(0.024)			
Births per 1,000 women aged 15–49	0.129	0.000	51.16	7,337
	(0.430)			
Log births per 1,000 women aged 15–49	0.003	0.000	3.93	7,337
	(0.008)			
Births (count)	-2.7	0.000	1,987.65	7,337
	(79.2)			

Notes: Standard errors clustered at the city level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Under- versus better-served cities, average of the post-2006 event-study coefficients, 2006 omitted (2003 in Panel B, the last pre-policy year with registered births). Panel A is every woman aged 25 to 45, mothers or not, so the sample does not depend on having a child; children are those living in the household. Panel B is the vital statistics register, available for 2000, 2003, 2010–2012 and 2014–2020, so its pre-period test compares only 2000 with 2003. Panel C is the INE projection of the population under one year old, a modelled estimate re-anchored at each census, which is why every pre-period test rejects. [a]Counts every child living in the household, so it moves with household composition and is not the woman's own fertility.

F Proofs of the Model

Throughout, I drop the subscript $t = 2$, since all the implications are about the second period, and I write $m = y + wh$ for the income of the household before paying for care. Since the increase in $\pi_{J,2}$ is not anticipated, the decisions of the first period, and so the wage w_2 of each mother, do not change with the policy. Also, the JUNJI draw is independent of the characteristics of the mother, so the policy only changes the choice set of the mothers who get a place.

Allocations. With external care $j \neq H$ and hours h, the mother sets $b = \bar{b}$ and divides the rest of her time between leisure and household work in fixed proportions, while at home she divides all her time L among leisure, household work and her child. With $\Phi = \phi_l + \phi_z + \phi_b + \phi_q$ and $\bar{q}_j = \left[\alpha(\theta_j \bar{a})^\rho + (1-\alpha)\bar{b}^\rho\right]^{1/\rho}$,

$$l^h = \frac{\phi_l\,(L - h - \bar{b})}{\phi_l + \phi_z}, \qquad z^h = \frac{\phi_z\,(L - h - \bar{b})}{\phi_l + \phi_z}, \tag{18}$$

$$V_j(h) = \ln(y - p_j + wh) + \phi_l \ln l^h + \phi_z \ln z^h + \phi_b \ln \bar{b} + \phi_q \ln \bar{q}_j, \tag{19}$$

$$l^H = \frac{\phi_l L}{\Phi}, \qquad z^H = \frac{\phi_z L}{\Phi}, \qquad b^H = \frac{(\phi_b + \phi_q) L}{\Phi}, \tag{20}$$

where l^1, z^1 (l^0, z^0) denote the allocations with $h = \bar{h}$ ($h = 0$). The wage at which working with provider j is as good as staying at home is

$$w_j^{**} = \frac{y\,e^{-\Delta_j} - (y - p_j)}{\bar{h}}, \qquad \Delta_j = \phi_l \ln \frac{l^1}{l^H} + \phi_z \ln \frac{z^1}{z^H} + \phi_b \ln \frac{\bar{b}}{b^H} + \phi_q \ln \frac{\bar{q}_j}{q^H}. \tag{21}$$

Values. If the child is in care option $j \neq H$, by the assumption on maternal time of Section II the mother sets $b = \bar{b} = S - \bar{a}$ and divides $L - h - \bar{b}$ between leisure and household work, which gives l^h and z^h in (18) and the value $V_j(h)$ in (19). If the child is at home, $h = 0$, $x = y$ and $q = (1-\alpha)^{1/\rho} b$, so the mother maximizes $\ln y + \phi_l \ln l + \phi_z \ln z + (\phi_b + \phi_q) \ln b$ subject to $l + z + b = L$, which gives the shares in (20) and

$$V_H = \ln y + \phi_l \ln l^H + \phi_z \ln z^H + \phi_b \ln b^H + \phi_q \ln q^H, \qquad q^H = (1-\alpha)^{1/\rho} b^H.$$

The development of the child $\bar{q}_j$ is increasing in θ_j, so (6) gives $\bar{q}_I < \bar{q}_M < \bar{q}_J < \bar{q}_P$.

Reservation wages. For $j \neq H$, $V_j(\bar{h}) \geq V_j(0)$ if and only if

$$\ln(y - p_j + w\bar{h}) - \ln(y - p_j) \geq \phi_l \ln \frac{l^0}{l^1} + \phi_z \ln \frac{z^0}{z^1} = (\phi_l + \phi_z) \ln \left(\frac{l^0 + z^0}{l^1 + z^1} \right) = \Gamma,$$

where the middle equality uses that leisure and household work keep the same proportions, so $l^0/l^1 = z^0/z^1 = (l^0 + z^0)/(l^1 + z^1)$. This gives $w \geq w_j^*$ in (8). Since w_j^* is increasing in $y - p_j$, (6) gives

$$w_P^* < w_M^* < w_J^* < w_I^*.$$

For the second reservation wage, subtracting V_H from $V_j(\bar{h})$ term by term gives

$$V_j(\bar{h}) - V_H = \ln \left(\frac{y - p_j + w\bar{h}}{y} \right) + \Delta_j,$$

with Δ_j as in (21). Δ_j does not depend on w, and θ_j enters only through its last term, so Δ_j is increasing in θ_j. The third term is negative because $\bar{b} < b^H$, and the first two are negative because $l^1 + z^1 < l^H + z^H$, both by the assumption on maternal time of Section II. Then $V_j(\bar{h}) \geq V_H$ if and only if $\ln(y - p_j + w\bar{h}) \geq \ln y - \Delta_j$, which gives $w \geq w_j^{**}$. Since $p_J < p_M$ and $\Delta_J > \Delta_M$,

$$w_M^{**} - w_J^{**} = \frac{y(e^{-\Delta_M} - e^{-\Delta_J}) + p_M - p_J}{\bar{h}} > 0,$$

and this difference goes to zero as $\theta_M \to \theta_J$ and $p_M \to p_J$. Both reservation wages are increasing in y.

Enrollment. Let s_J be the share of mothers who choose JUNJI when they have a place. Since the draw is independent of the characteristics of the mother, JUNJI enrollment is $\pi_{J,2}\, s_J$, and its derivative with respect to $\pi_{J,2}$ is $s_J \geq 0$. By part (a) of the substitution result below, every mother who uses a municipal daycare chooses JUNJI when she has a place, so $s_J > 0$ as long as some mothers use municipal daycares. □

Substitution, equation (9). (a) For any h,

$$V_J(h) - V_M(h) = \ln \left(\frac{m - p_J}{m - p_M} \right) + \phi_q \ln \left(\frac{\bar{q}_J}{\bar{q}_M} \right) > 0,$$

since both terms are positive. If (M, h) was the best option before, then $V_J(h) > V_M(h) \geq V_k(h')$ for every other option k and hours h', so the best option with a JUNJI place uses JUNJI. The hours of the mother may change, which is part (b) of the employment result below.

(b) Holding h fixed, $V_J(h) - V_I(h) = \ln(1 - p_J/m) + \phi_q \ln(\bar{q}_J/\bar{q}_I)$, which is positive under the

condition in the proposition. Then $V_J(h) > V_I(h) \geq V_k(h')$ for every other option, and the mother moves to JUNJI. The condition is sufficient, and it is also necessary when the mother does not change her hours.

(c) Holding h fixed, define $D(m) = V_P(h) - V_J(h) = \ln\big((m-p_P)/(m-p_J)\big) + \phi_q \ln(\bar{q}_P/\bar{q}_J)$ for $m > p_P$. Its derivative is

$$D'(m) = \frac{1}{m-p_P} - \frac{1}{m-p_J} = \frac{p_P - p_J}{(m-p_P)(m-p_J)} > 0,$$

$D(m) \to -\infty$ when $m \to p_P$, and $D(m) \to \phi_q \ln(\bar{q}_P/\bar{q}_J) > 0$ when $m \to \infty$. So there is a unique $\bar{m}$ with $D(\bar{m}) = 0$, and a mother with $m > \bar{m}$ keeps the private daycare. □

Employment, equation (10). I look at each group of mothers, depending on what they were doing before getting a JUNJI place. By the substitution result, a mother who gets a place and moves to JUNJI chooses $(J, \bar{h})$ over $(J, 0)$ if and only if $w \geq w_J^*$.

Mothers who were not working. A mother at $(k, 0)$ with $k \in \{M, P\}$ had $V_k(0) \geq V_k(\bar{h})$, so $w < w_k^* < w_J^*$, and she does not start working. A mother at $(I, 0)$ had $w < w_I^*$, so she starts working only if $w_J^* \leq w < w_I^*$, an interval of width $p_J(e^\Gamma - 1)/\bar{h}$. A mother at $(H, 0)$ had $V_H \geq V_M(\bar{h})$, so $w < w_M^{**}$. To start working with JUNJI she needs $V_J(\bar{h}) \geq V_H$ and $V_J(\bar{h}) \geq V_J(0)$, meaning $w \geq w_J^{**}$ and $w \geq w_J^*$. So the mothers who enter have $\max\{w_J^{**}, w_J^*\} \leq w < w_M^{**}$, which is part (a). In particular, a mother who was not working and has $w < w_J^{**}$ or $w < w_J^*$ does not start working.

Mothers who were working. A mother at $(M, \bar{h})$ had $w \geq w_M^*$. She moves to JUNJI, and she stops working if $w < w_J^*$. So the mothers who leave have $w_M^* \leq w < w_J^*$, an interval of width

$$w_J^* - w_M^* = \frac{(p_M - p_J)(e^\Gamma - 1)}{\bar{h}},$$

which is part (b). A mother at $(P, \bar{h})$ who moves to JUNJI (which requires $m < \bar{m}$) stops working if $w_P^* \leq w < w_J^*$. A mother at $(I, \bar{h})$ has $w \geq w_I^* > w_J^*$, so if she moves to JUNJI she keeps working.

The change in participation is the change in $\pi_{J,2}$ times the share of mothers who enter minus the share who leave, which has no definite sign. The intervals in (a) and in (b) for municipal mothers go to zero as $\theta_M \to \theta_J$ and $p_M \to p_J$. □

Sending the child to JUNJI without working. Subtracting term by term,

$$V_J(0) - V_H = \ln\left(\frac{y - p_J}{y}\right) + \phi_l \ln\frac{l^0}{l^H} + \phi_z \ln\frac{z^0}{z^H} + \phi_b \ln\frac{\bar{b}}{b^H} + \phi_q \ln\frac{\bar{q}_J}{q^H},$$

and the condition is that this difference is positive. The right-hand side is increasing in θ_J, since $\bar{q}_J$ is. Writing $D_q = \alpha(\theta_J \bar{a})^\rho + (1-\alpha)\bar{b}^\rho$, and since l^0, z^0, l^H, z^H and b^H do not depend on α, its

derivative with respect to α is

$$\frac{\phi_q}{\rho}\left[\frac{(\theta_J\bar{a})^\rho-\bar{b}^\rho}{D_q}+\frac{1}{1-\alpha}\right]=\frac{\phi_q}{\rho}\frac{(\theta_J\bar{a})^\rho}{(1-\alpha)D_q}>0,$$

so the condition is easier to satisfy when the weight of the daycare in the development of the child is higher. □

Reported reasons. By the reporting rule, the share that reports childcare is the share at $(H,0)$, and the share that reports household chores or no interest is the share at $(j,0)$ with $j\neq H$. A mother at $(H,0)$ who gets a place either stays at home, starts working with JUNJI (part (a) of the employment result), or sends her child to JUNJI without working (previous paragraph), so the share that reports childcare decreases. The mothers who send their child to JUNJI without working add to the share that reports chores or no interest, and so do the working mothers who stop working in part (b) of the employment result. The only mothers who leave this group are the mothers at $(I,0)$ who start working, in the small interval of part (a) of the employment result. So the share that reports chores or no interest increases as long as the mothers who send their child to JUNJI without working are more than the mothers in that interval. □

Nonmarket time, equation (11). A mother at $(H,0)$ has $z^H=\phi_z L/\Phi$. If she moves to $(J,0)$, she has $z^0=\phi_z(L-\bar{b})/(\phi_l+\phi_z)$. Then

$$z^0-z^H=\phi_z\left[\frac{L-\bar{b}}{\phi_l+\phi_z}-\frac{L}{\Phi}\right]=\frac{\phi_z}{\phi_l+\phi_z}\left[\frac{(\phi_b+\phi_q)L}{\Phi}-\bar{b}\right]=\frac{\phi_z}{\phi_l+\phi_z}\left(b^H-\bar{b}\right),$$

which is positive by the assumption on maternal time of Section II. The same steps with ϕ_l give the change in leisure. If the mother moves between two options $j,k\neq H$ with the same h, both have $b=\bar{b}$ and the same $L-h-\bar{b}$, so her time use does not change. □

G How Close are JUNJI and Municipal Daycares

The substitution results suggest that families moved their children from municipal care to JUNJI. One reason could be that the new JUNJI daycares were closer to their homes. I do not observe the address of the households or the daycares, so I approximate how close each provider is with a simple measure. For each city and provider, I compute the radius that each facility would serve if the city were divided equally among them, $r = \sqrt{A/(\pi n)}$, where A is the area of the city and n the number of facilities. If the facilities were placed at random, the expected distance from a house to the nearest one would be $0.89\,r$. I use the 40 cities with at least 1,000 inhabitants per km^2, where the area of the city is close to the area where people live.

In 2006, the nearest JUNJI daycare was farther than the nearest municipal school in most dense cities (Table 22). In the under-served cities the median JUNJI radius was 1.19 km against 0.91 km for municipal schools, and JUNJI was closer only in 12% of them. By 2024 the JUNJI radius fell to 0.66 km while the municipal one stayed around one kilometer, and JUNJI is closer in 53% of these cities; in better-served cities the share goes from 25% to 50%. Figure 35 shows the same change for the metropolitan region of Santiago. The expansion, then, moved JUNJI from being the farther public option to being the closer one in many cities, which is consistent with families moving their children from municipal care. I use this only as descriptive support, since I do not observe where each family lives.

Table 22: Distance to the nearest JUNJI and municipal facility: radius served by each facility

		Facilities		Radius (km)		JUNJI
	Cities	JUNJI	Municipal	JUNJI	Municipal	closer (%)
Panel A. Medians, cities with at least 1,000 inhabitants per km^2						
Under-served, 2006	32	6	12	1.19	0.91	12
Under-served, 2024	32	14	11	0.66	0.99	53
Better-served, 2006	8	10	12	0.99	0.89	25
Better-served, 2024	8	19	13	0.76	0.94	50
Panel B. Examples						
Maipú, 2006		10	20	2.08	1.47	
Maipú, 2024		36	18	1.09	1.55	
Quilicura, 2006		5	8	1.92	1.52	
Quilicura, 2024		22	8	0.92	1.52	
Puente Alto, 2006		27	20	1.02	1.18	
Puente Alto, 2024		50	19	0.75	1.21	
Santiago, 2006		2	19	1.92	0.62	
Santiago, 2024		25	22	0.54	0.58	

Notes: The radius is $r = \sqrt{A/(\pi n)}$, where A is the area of the city and n the number of facilities of each provider: the radius of the circle each facility would serve if the city were divided equally among them. If facilities were placed at random, the expected distance to the nearest one would be $0.89\,r$. JUNJI counts care-providing facilities; municipal counts municipal schools that offer early education, which are mostly pre-kindergarten. Only cities with at least 1,000 inhabitants per km^2 in 2017 are used, because only there is the area of the city close to the area where people live. *JUNJI closer* is the share of cities where the JUNJI radius is smaller than the municipal one. Area and 2017 population of each city from the public comunas table in github.com/altazor-1967/Comunas-de-Chile.

Figure 35: Municipal radius minus JUNJI radius, Santiago metropolitan region, 2006 and 2024

